



**TRAVEL AGENCY WEBSITE WITH PERSONALIZED
TRAVEL PROGRAM RECOMMENDATION BASED ON
CLUSTERING ALGORITHM**

**NATTAPAT CHANTRACHIRATHUMRONG
THANAYTHON KINGRAT
PHUCHIT YUTTAGLA
PHUSANISA SUKSOPHONTHANA
WASAWAT SRIHAWONG**

**BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

**MAE FAH LUANG UNIVERSITY
ACADEMIC YEAR 2023**

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**A SENIOR PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

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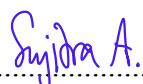
TRAVEL AGENCY WEBSITE WITH PERSONALIZED TRAVEL PROGRAM RECOMMENDATION BASED ON CLUSTERING ALGORITHM

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**THIS SENIOR PROJECT HAS BEEN APPROVED
TO BE A PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING
ACADEMIC YEAR 2023**

EXAMINING COMMITTEE

.....  **ADVISOR**
(Asst. Prof. Pattaramon Vuttipittayamongkol, Ph.D.)

.....  **COMMITTEE**
(Asst. Prof. Sujitra Arwatchananukul)

.....  **COMMITTEE**
(Dr. Patcharaporn Panwong)

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NATTAPAT CHANTRACHIRATHUMRONG
THANAYTHON KINGRAT
PHUCHIT YUTTAGLA
PHUSANISA SUKSOPHONTHANA
WASAWAT SRIHAWONG

Title	Web Application Tourist Guide Based On Clustering Algorithm	
Author	Mr. Nattapat Chantrachirathumrong Mr. Thanaython Kingrat Mr. Phuchit Yuttagla Ms. Phusanisa Suksophonthana Mr. Wasawat Srihawong	
Degree	Bachelor of Engineering (Computer Engineering)	
Supervisory Committee	Asst. Prof. Pattaramon Vuttipittayamongkol, Ph.D. Asst. Prof. Sujitra Arwatchananukul Dr. Patcharaporn Panwong	Advisor Committee Committee

Abstract

This project proposes the development of a web application that serves as a tourist guide for visitors to a tourist place. The application uses a clustering algorithm to group and categorize popular clients' attractions based on their attributes and proximity to one another. Upon completing the self-administered questionnaire, users have the option to select a personalized travel itinerary based on the recommended tour package generated from the predicted results. Includes recommended routes and estimated travel times between attractions. The main aim of the project is to make the traveler experience more efficient and enjoyable by reducing the time spent planning and navigating. and highlighting the most relevant and interesting places to visit.

Keyword: Web application, Tourist guide, Clustering algorithm

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CHAPTER 1

INTRODUCTION

1.1 Background and Rationale

Technology has changed the game for the tourism industry by giving tourists access to a ton of information about their desired destinations. However, with so much information available online, tourists can sometimes feel overwhelmed, making it hard for them to make informed decisions. To help with this, a new web app tourist guide has been created that uses a clustering algorithm to offer tourists a more personalized and efficient experience.

This website tourist guide groups the travel behavior of the tourist, using clustering algorithms. These algorithms can sort tourist attractions based on location, type, popularity, and user ratings. This means the app can give tourists recommendations based on their personal preferences and interests, making their experience smoother and more tailored to them.

1.2 Objective

1.2.1 To obtain a clustering algorithm that can classify the tourist characteristics and analyze each group's characteristics.

1.2.2 To create a website that recommends the tour based on their travel behavior and personal information.

1.3 Scope

1.3.1 Data from the project “Using factor analyses to understand the post-pandemic travel behavior in domestic tourism through a questionnaire survey”[1] by Asst. Prof. Dr. Tosporn Arreeras contains 460 samples and data from data collection of 67 samples.

1.4 Methodology

1.4.1 Proposal preparation and defense

1.4.2 Literature review and requirement gathering

1.4.3 Data exploration and data analysis

1.4.4 Initial model building and testing

1.4.5 Design website (Front-end) and database

1.4.6 Senior project 1 document preparation

1.4.7 Senior Project 1 presentation

	Pre-project				Project					Duty
Plan/Month	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec	
Developing model										4,5
Model testing										4,5
Create website (Back-end)										1,2,3
Combine model to website										All
Senior Project 2 document preparation										All
Senior Project 2 presentation										All

1.6 Expected Result

1.6.1 Development of a clustering algorithm that can group tourists based on their characteristics.

1.6.2 Design and development of a website that recommends tours based on the travel behavior and personal information of the tourists.

1.7 Equipment

1.7.1 Data collection

- Google Forms

1.7.2 Machine learning

- Jupyter notebook
- Docker

1.7.3 Website

- Figma
- Visual Studio Code
- React
- Bootstrap 5
- Express
- Node.js

- Prisma
- Docker
- Flask

1.7.4 Database

- MySQL
- phpmyadmin

CHAPTER 2

LITERATURE REVIEW

2.1 Related Theory

Clustering algorithm is the unsupervised learning that deals with data structure partition in unknown areas [2]. The complete definition for clustering described as instances in the same cluster must be similar as much as possible and instances in the different clusters must be different as much as possible. The traditional clustering algorithms can be divided into 9 categories: Based on the partition, Based on the hierarchy, Based on fuzzy theory, Based on distribution, Based on density, Based on graph theory, Based on the grid, Based on fractal Theory, and Based on the model.

Silhouette coefficient is an evaluation indicator of clustering algorithms [3]. Which evaluates the cluster based on the measure of the average distance between one data point and other data points in the same clusters and the average distance among different clusters. The silhouette value can vary between -1 and 1, with higher values indicating better-defined clusters and lower values indicating poorly defined clusters.

2.2 Related Research

1. The present study [4] focuses on tourism in Phra Nakhon Si Ayutthaya province, a popular destination among both domestic and international travelers. The study categorizes and manages the activities of those visitors who have entered the province, including purchasing souvenirs, dining at restaurants, and staying in hotels, to provide the best possible products and services. The study employs Hierarchical and K-Means algorithms in Two-step clustering analysis to classify visitors into four clusters. The clusters include senior tours with organizational trips (15.65% of total visitors), elderly tours with family (37.78% of total visitors), employees (23.58% of total visitors), and tourism lovers (23.01% of total visitors). The study also predicts travel behavior using six classification algorithms, namely J48, Naive Bayes, Bayes Network, Ripper, Logistic Regression, and K-nearest neighbors, to plan marketing strategies and improve products and services. The Bayes Network and Naive Bayes algorithms are found to be more effective in predicting travel behavior. The study concludes that the maximum distance is between Cluster 1 and Cluster 2.

Review : The present study employs a two-step clustering approach, which leads to higher accuracy in analyzing each group's tourism behavior, including predictions. However, it is crucial to note that this research is specific to Ayutthaya province. Therefore, the resulting model is applicable only within this province, and not every entrepreneur may choose to use it to enhance their business. Consequently, only a subset of businesses may utilize this model, and it remains unclear whether it is effective or not.

2. This research project [5] includes three distinct login categories: Admin, Agents, and Users. Admin login grants privileged access for agent approval and provides the ability to view user-generated package reviews. Agent login allows the privilege to add destinations, view bookings, and access booking history. Users, on the other hand, have the privilege to view popular destinations and search for new destinations based on predictive ratings and reviews.

Review : For this study on Intelligent Tour Operators, data mining was employed to suggest the optimal travel guide for users. The recommendation is made by predicting the user's interest in the search term, utilizing a search engine to look for relevant packages provided by the tour guide with the best features. However, this study did not incorporate a machine learning model that could provide the most accurate grouping of results. Hence, it is necessary to rely on agents to advise users on the best travel guides.

2.3 Related Technology

Identify both hardware and software required for the project as well as their references. The details should include the target version and you could explain each item separately as a new subtopic.

2.3.1 Machine learning

- K-Means

K-Means is one of the simplest unsupervised learning algorithms based on partitions that solve the well known clustering problem . The main idea is to define k centroids, one for each cluster and take each point belonging to a given data set and associate it to the nearest centroid. Then re-calculate k new centroids as centers of the clusters resulting from the previous step. After these k new centroids, a new binding has to be done between the same data points and the nearest new centroid. A loop has been generated. As a result of this loop it may notice that the k centroids change their location step by step until no more changes are done.

- Agglomerative hierarchical clustering

Agglomerative hierarchical clustering is hierarchy-based clustering since it builds a binary merge tree starting from leaves that contain data elements to the root that contains the full data-set. The graphical representation of that tree that embeds the nodes on the plane is called a dendrogram. To implement a hierarchical clustering algorithm, one has to choose a linkage function that defines the distance between any two subsets . A hierarchical clustering is monotonous if and only if the similarity decreases along the path from any leaf to the root, otherwise there exists at least one inversion.

- **Docker**

Docker takes away repetitive, mundane configuration tasks and is used throughout the development lifecycle for fast, easy and portable application development – desktop and cloud. Docker’s comprehensive end to end platform includes UIs, CLIs, APIs and security that are engineered to work together across the entire application delivery lifecycle.

2.3.2 Website

- **Figma**

Figma is a cloud-based design and prototyping tool used to create user interfaces, websites, mobile apps, and other digital products. It is a collaborative platform that enables teams to work together on design projects in real-time from anywhere in the world.

- **Visual Studio Code**

Visual Studio Code is a free, open-source code editor developed by Microsoft. It is a lightweight, cross-platform tool that is designed for web and application developers, and provides a range of features to support coding, debugging, and version control.

- **React**

React is a popular open-source JavaScript library for building user interfaces. It was developed by Facebook and is now maintained by Facebook and a large community of developers.

- **Bootstrap 5**

Bootstrap 5 is an open-source front-end web development framework that allows developers to create responsive and mobile-first web applications. It is the most recent version of the Bootstrap framework and comes equipped with a variety of new features and improvements, such as an updated responsive grid system, updated CSS styles, increased customizability, and updated utilities.

- **HTML**

HTML (Hypertext Markup Language) is the standard markup language for creating and structuring content on the World Wide Web. It uses tags to define elements such as headings, paragraphs, links, images, and more. HTML provides the basic structure for web pages and is essential for building and designing websites.

- **CSS**

CSS (Cascading Style Sheets) is a language that styles HTML or XML documents, controlling their presentation and layout. It separates content and design, enabling consistent styling across web pages through selectors, properties, and values.

- **JavaScript**

JavaScript is a programming language commonly used for web development. It enables dynamic, interactive behavior in web browsers, allowing developers to create features like animations, form validation, and real-time updates. As a client-side scripting language, JavaScript runs in the browser, complementing HTML and CSS to enhance the functionality and user experience of websites.

- **Node.js**

Node.js is an open-source, server-side JavaScript runtime environment that enables the execution of JavaScript code outside a web browser. It is designed for building scalable and high-performance network applications, making it popular for creating server-side applications and APIs.

- **Python**

Python is a versatile, high-level programming language known for its readability and simplicity. Widely used in various domains, including web development, data science, and artificial intelligence, Python emphasizes code readability and ease of use, making it a popular choice for both beginners and experienced developers.

- **Prisma**

Prisma JS typically refers to Prisma, an open-source database toolkit for Node.js and TypeScript. Prisma simplifies database access by providing a type-safe and auto-generated query builder.

- **Express**

Express.js is a minimalistic web application framework for Node.js. It simplifies server-side development with features like routing, middleware support, and template engines.

- **Docker**

Docker serves as an open platform for the development, shipping, and execution of applications, allowing the separation of applications from infrastructure for swift software delivery. It enables the management of infrastructure in a manner similar to application management.

- **Flask**

Flask, a micro web framework in Python, is categorized as such due to its independence from specific tools or libraries. It lacks built-in components like a database abstraction layer or form validation, relying on external third-party libraries for common functionalities.

2.3.3 Database

- **MySQL**

MySQL is an open-source relational database management system (RDBMS) that uses structured query language (SQL) for managing and organizing data. MySQL also provides a range of features for managing data, including data manipulation, indexing, and transaction processing, making it a powerful tool for managing large amounts of data.

- **PhpMyAdmin**

PhpMyAdmin is a web-based tool for managing MySQL and MariaDB databases. It offers a user-friendly interface for tasks like database creation, SQL query execution, user management, and data import/export. With features for table maintenance and server status monitoring, it simplifies database administration tasks, making it popular among developers and administrators.

CHAPTER 3

ANALYSIS AND DESIGN

3.1 Workflow Overview

The existing could be demonstrated by a use case diagram in Figure 3.1.

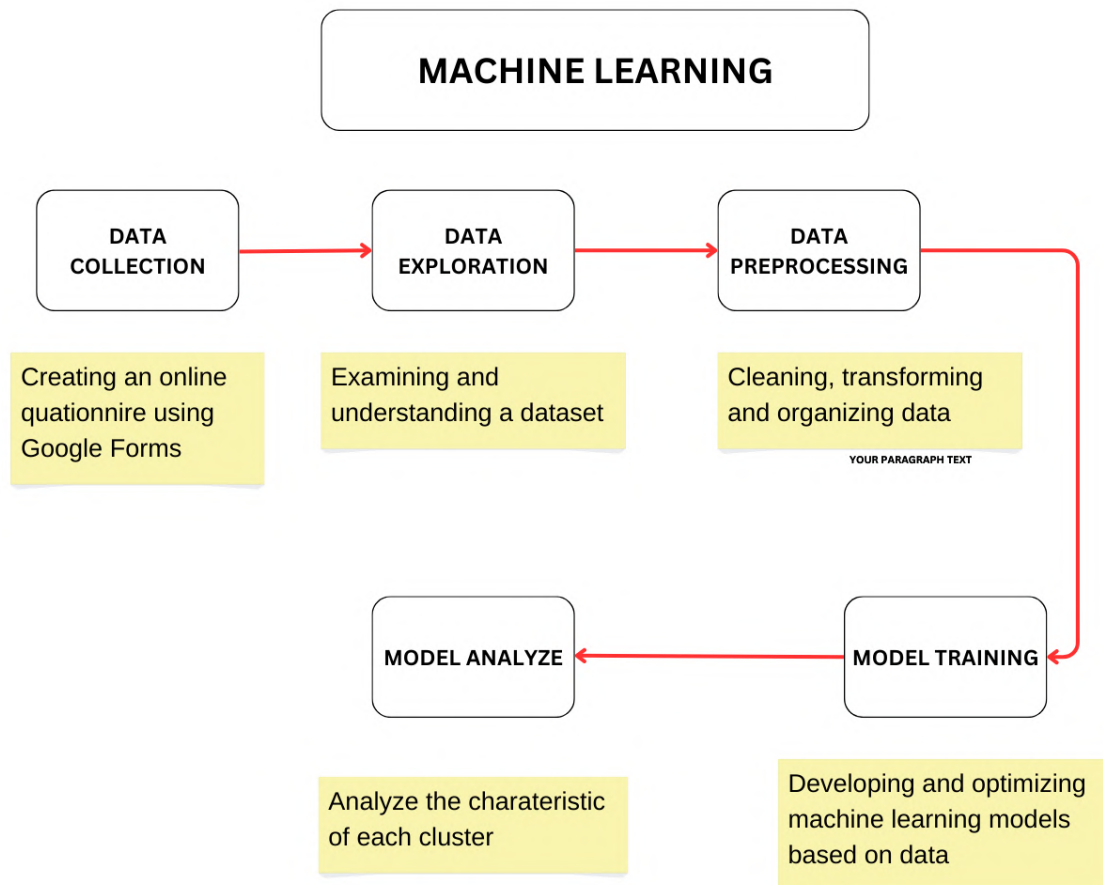


Figure 3.1 Workflow of Machine Learning

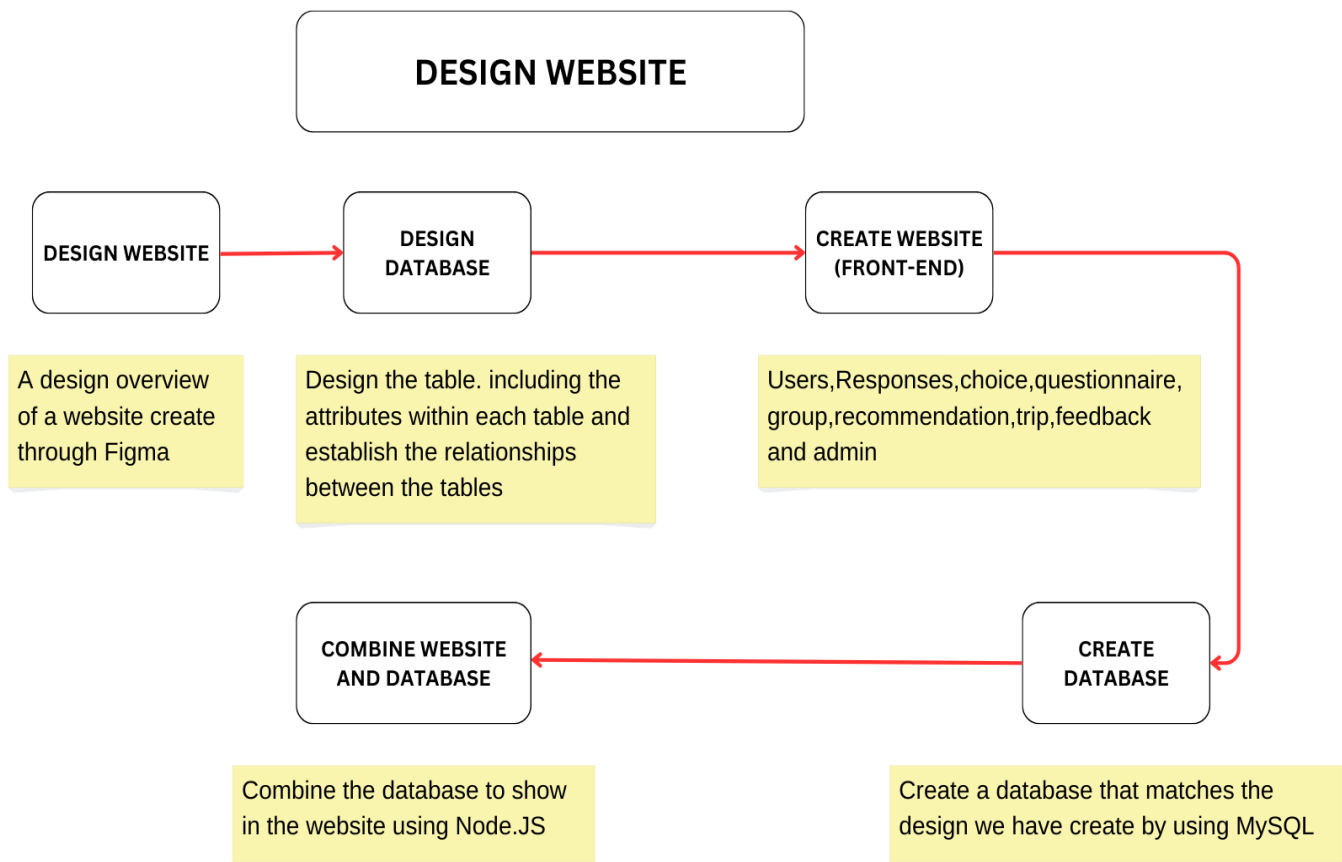


Figure 3.2 Workflow of Design Website

3.2 Dataset

This dataset is obtained from a research project “ Travel questionnaire before COVID-19 and after COVID-19”, with 527 people.

There are a total of 19 variables in Table 3.1

Table 3.1 Variable with Description

Order	Feature name	Values	Data type
1	gender	0: Male, 1: Alternative, 2: Female	Nominal
2	age	0: Less than or equal to 20 years old, 1: 21-30 years old, 2: 31-40 years old, 3.0: 41-50 years old, 4: 51-60 years old, 5: Over 60 years old	Ordinal
3	education	0: Under bachelor's degree, 1: Bachelor's degree or equivalent, 2: Higher bachelor's degree	Ordinal

Order	Feature name	Values	Data type
4	occupation	0: Government officers, 1: Business owner, 2: Student, 3: Private company employees, 4: Housewife/Househusband, 5: Freelancer, 6: Government employee, 7: Others, 8: Farmer	Nominal
5	income	0: Less than 15,000 THB, 1: 15,000 - 30,000 THB, 2: 30,001 - 45,000 THB, 3: 45,001 - 60,000 THB, 4: More than 60,000 THB	Ordinal
6	per_vehicle	0: Other, 1: Private car, 2: None, 3: Motorcycle	Nominal
7	household	0: 1 person, 1: 2-3 people, 2: 4-5 people, 3: More than 5 people	Ordinal
8	status	0: Married, 1: Single, 2: Not mentioned	Nominal
9	per_region	0: Central, 1: Western, 2: Eastern, 3: North-East, 4: North, 5: South	Nominal
10	after_freq	0: Less than 2 times/year, 1: 2-3 times/year, 2: 4-5 times/year, 3: More than 5 times/year	Ordinal
11	after_season	0: Rainy season, 1: Hot season, 2: Cold season	Nominal
12	after_region	0: Central, 1: Western, 2: Eastern, 3: North-East, 4: North, 5: South	Nominal
13	after_person	0: Single, 1: 1-2 persons, 2: 3-4 persons, 3: 5 persons, 4: More than 5 persons	Ordinal
14	after_vehicle	0: Motorcycle, 1: Private car, 2: Public transport, 3: Train, 4: Airplane, 5: Passenger ship	Nominal
15	after_expenditure	0: Lower than 500 THB, 1: 501-1500 THB, 2: 1501-2500 THB, 3: 2501-3500 THB, 4: More than 3500 THB	Ordinal
16	after_day_travel	0: 1 day, 1: 1-2 days, 2: 3-4 days, 3: More than 4 days	Ordinal
17	after_period_time	0: Weekday (Monday-Friday), 1: Holidays, 2: Weekend (Saturday-Sunday)	Nominal
18	after_want_travel	0: I'd love to, 1: I'm not sure, 2: Definitely not!	Nominal
19	after_type	0: Environment destination (e.g. beach, mountain, waterfall), 1: Historical destination (e.g. temple, historical park, museum), 2: Cultural destination (e.g. floating market, traditional festival, cultural performance), 3: Recreational destination (e.g.	Nominal

Order	Feature name	Values	Data type
		entertainment venues, amusement parks, zoos, shopping malls, concerts), 4: Ethnic destination (e.g. Kariang village, coffee planting workshop, homestay), 5: Business destination (e.g. seminars, business negotiations, meetings)	

3.2.1 Data collection

Collect more data by using google form

แบบสอบถามการเดินทางท่องเที่ยวก่อนโควิด19และหลังโควิด19

6331501064@lamduan.mfu.ac.th [Switch account](#)

Not shared

* Indicates required question

ข้อมูลส่วนบุคคล

เพศ *

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อายุ *

☐ ต่ำกว่าหรือเท่ากับ 20 ปี

☐ 21-30 ปี

☐ 31-40 ปี

☐ 41-50 ปี

☐ 51-60 ปี

☐ มากกว่า 60 ปี

Figure 3.3 Show the google form that we collect data.

3.2.2 Data exploratory

Figure 3.4-3.22 shows the frequency of some features.

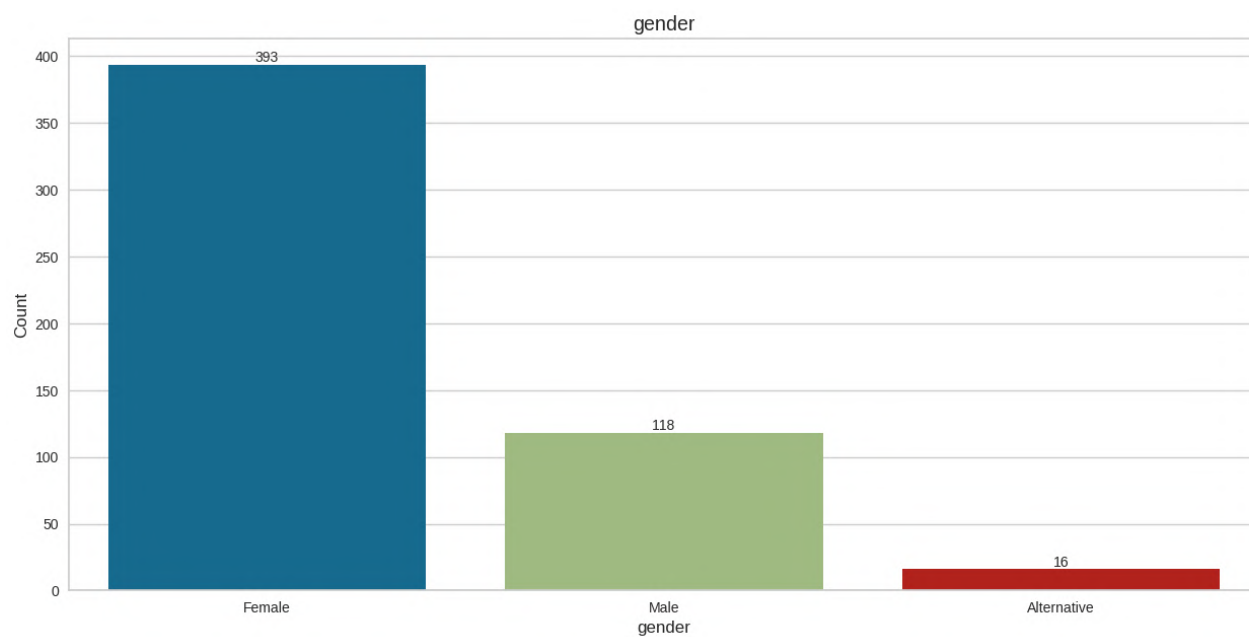


Figure 3.4 Gender

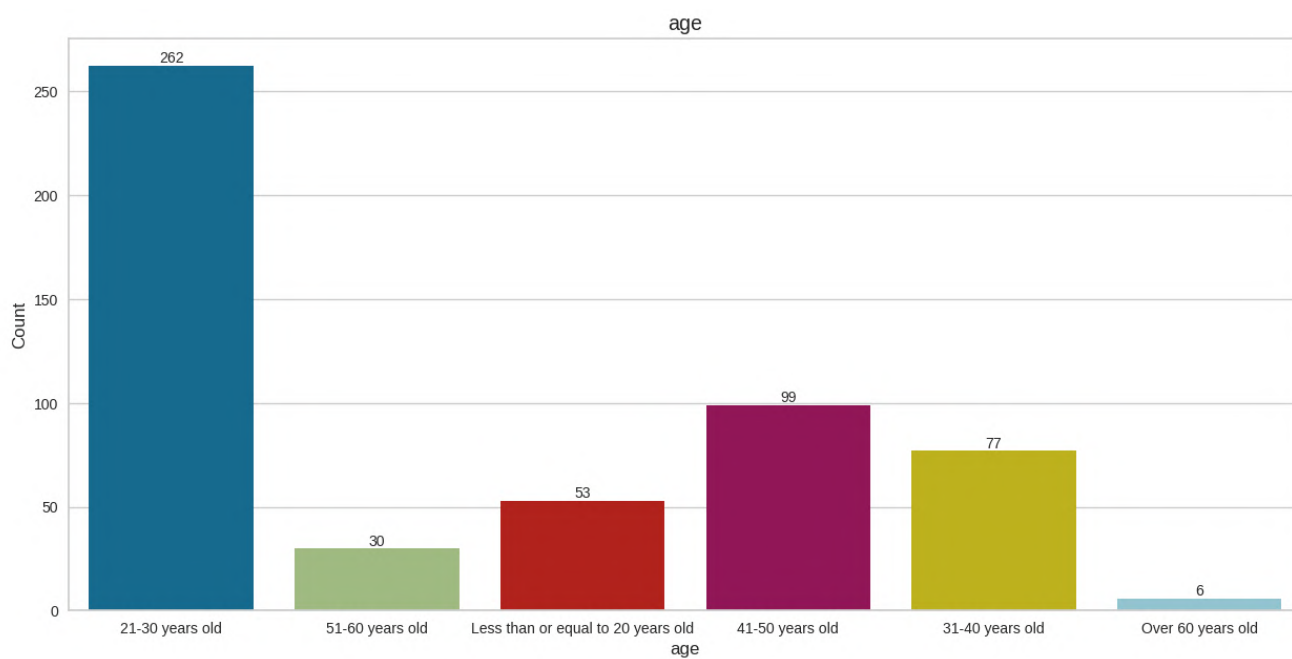
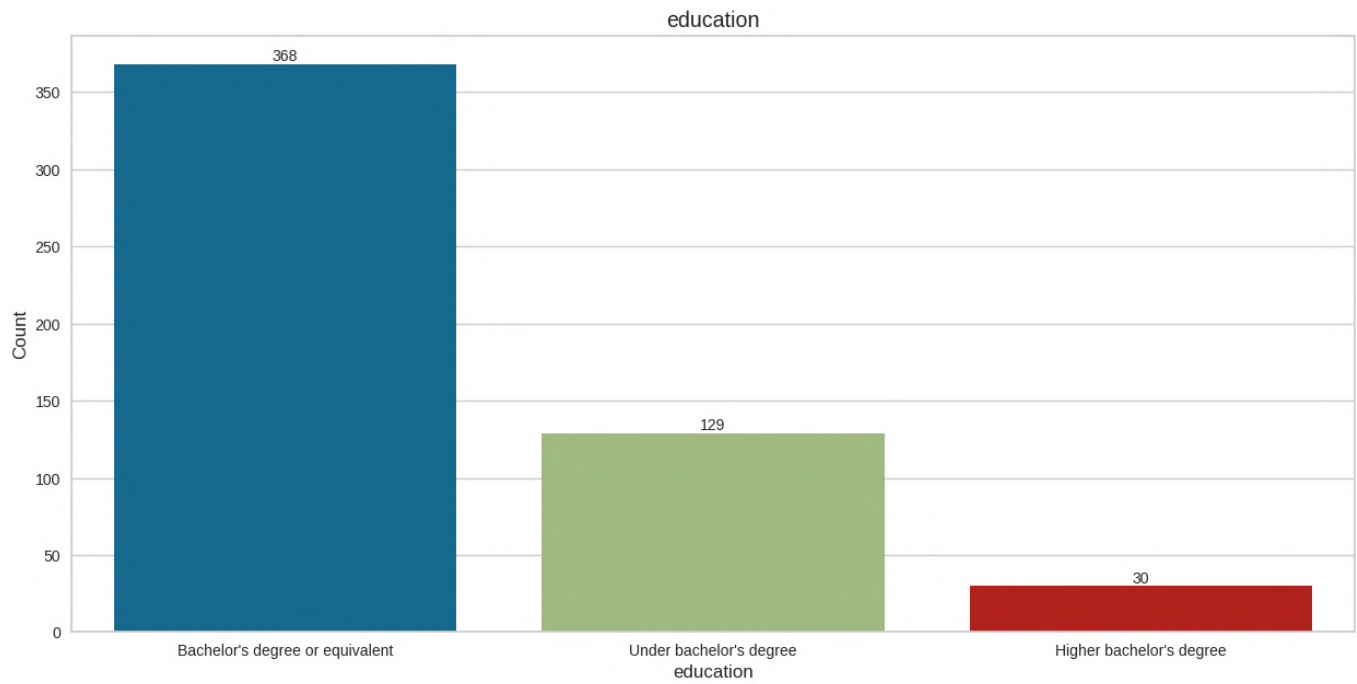
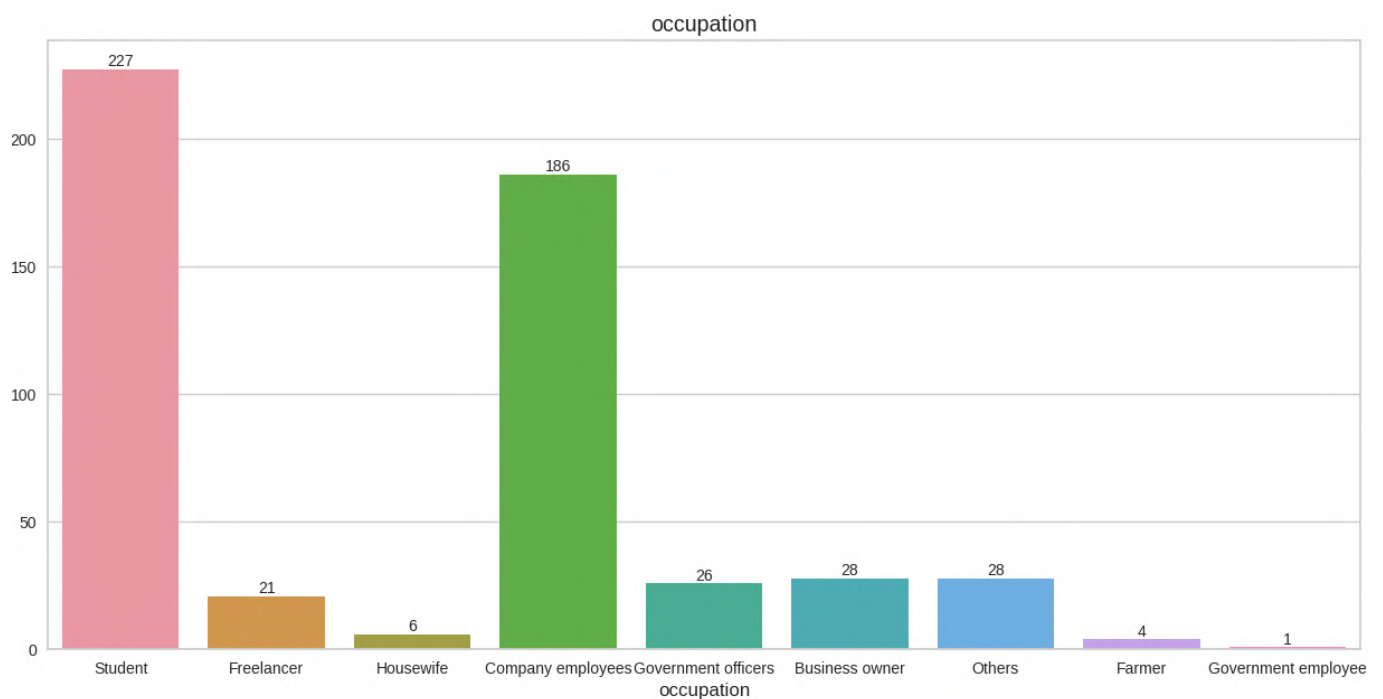


Figure 3.5 Age

**Figure 3.6 Education****Figure 3.7 Occupation**

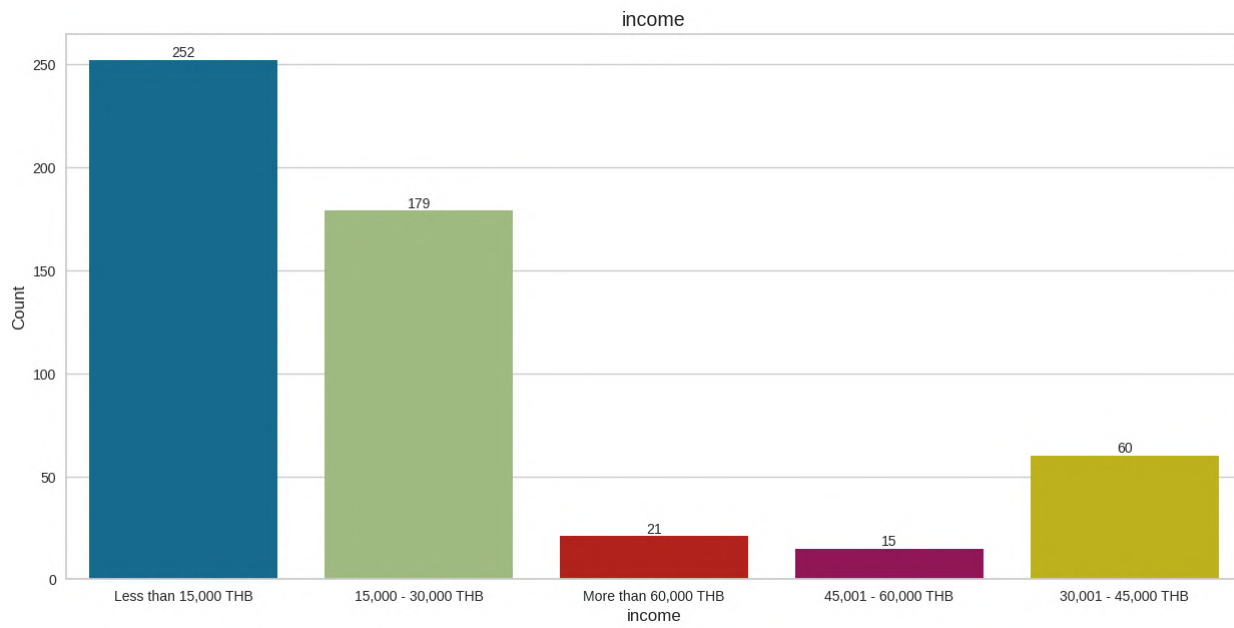


Figure 3.8 Income

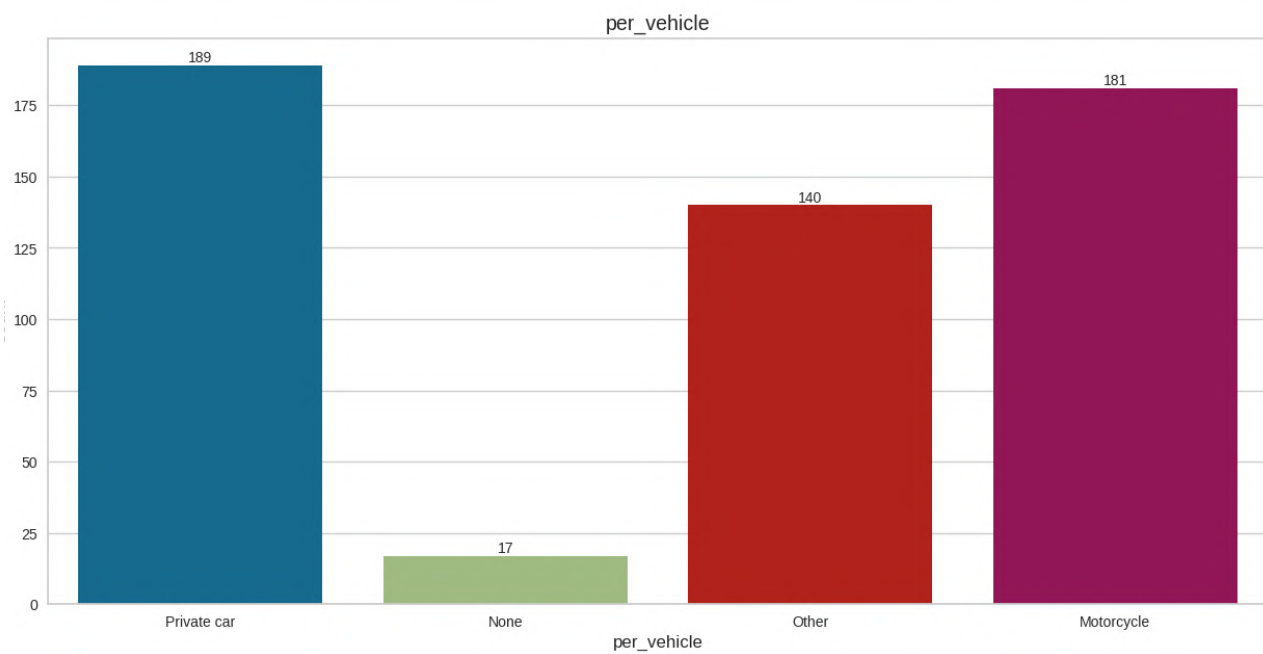


Figure 3.9 Per_vehicle

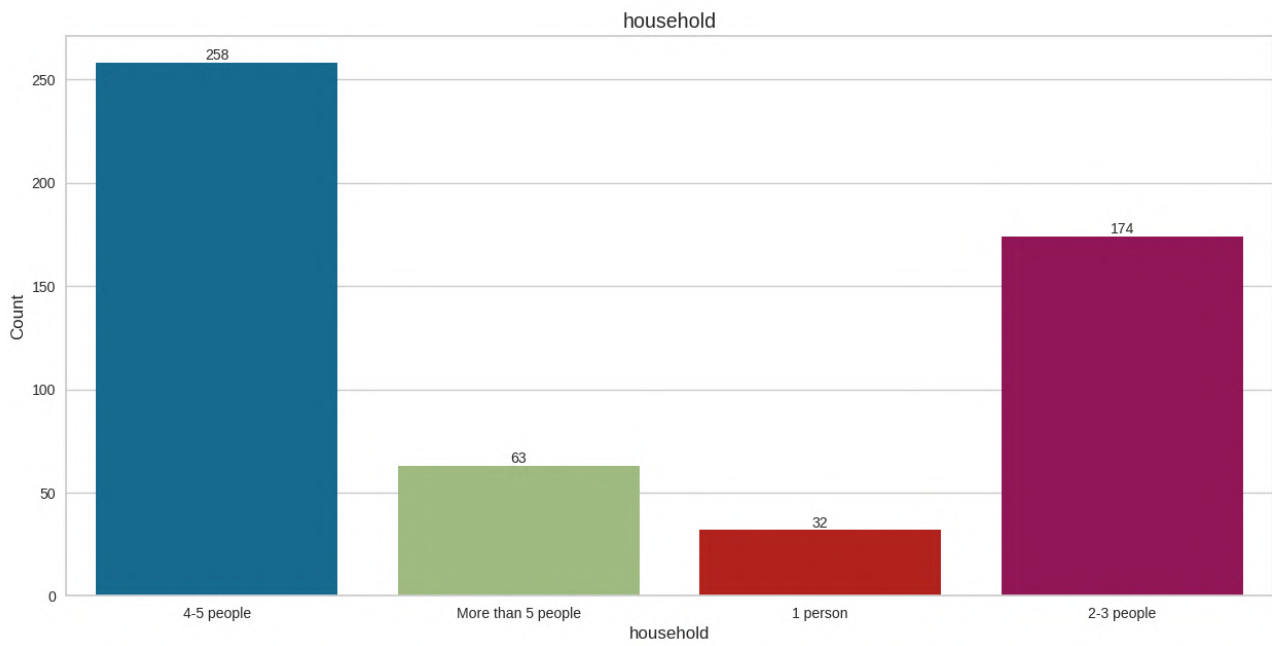


Figure 3.10 Household

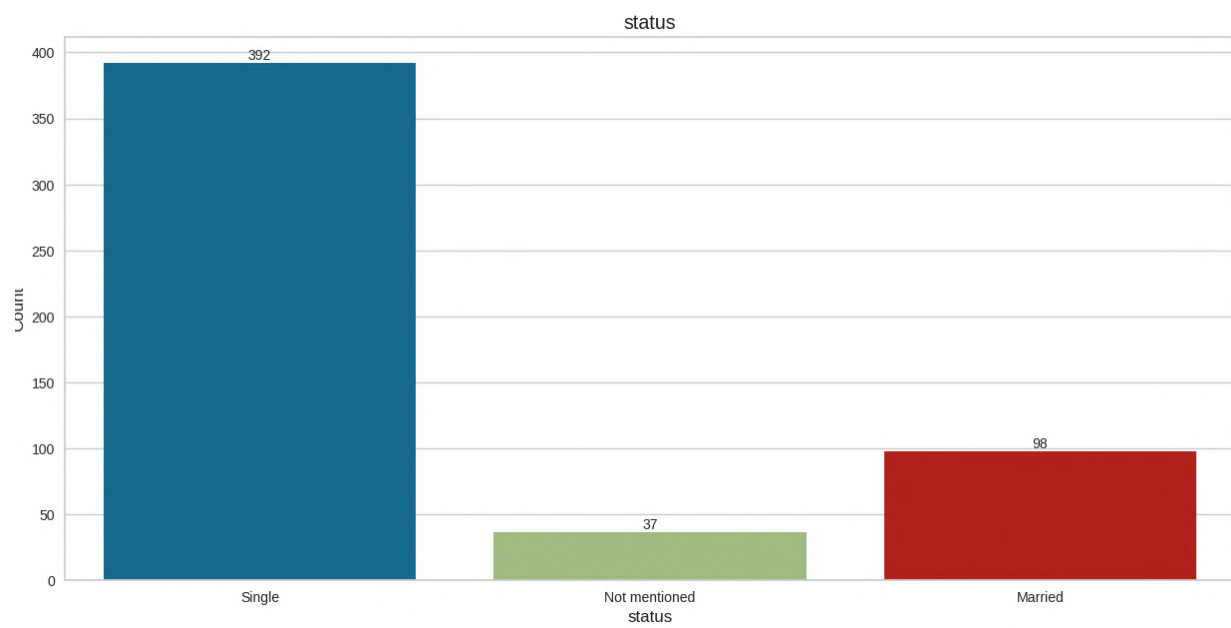


Figure 3.11 Status

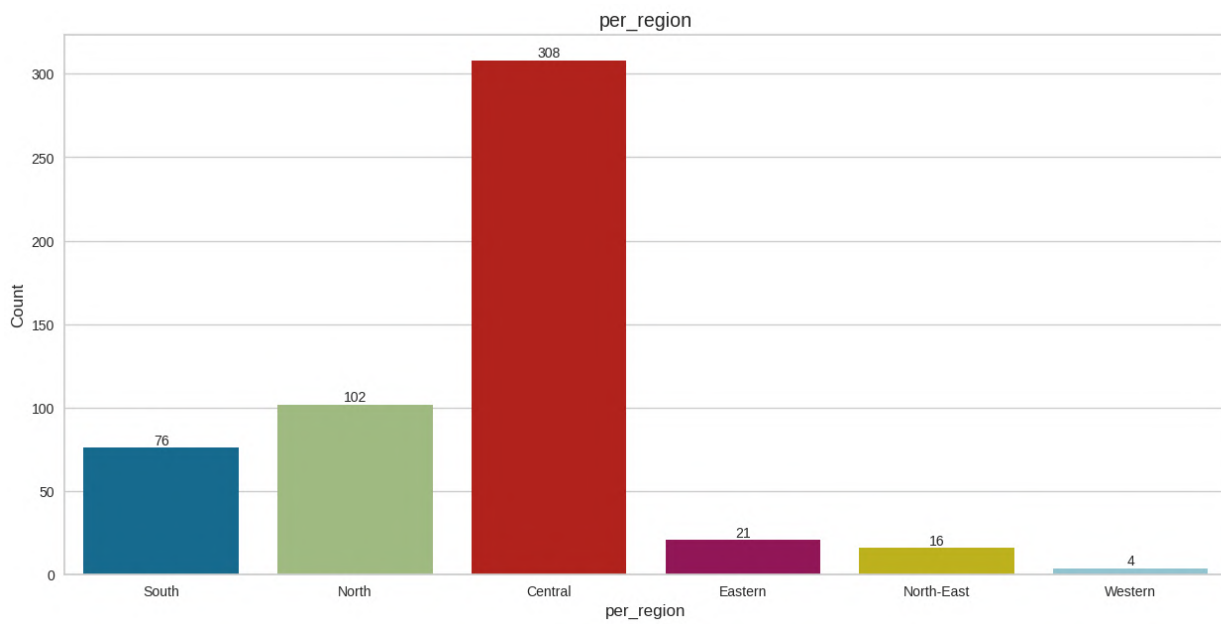


Figure 3.12 Per_region

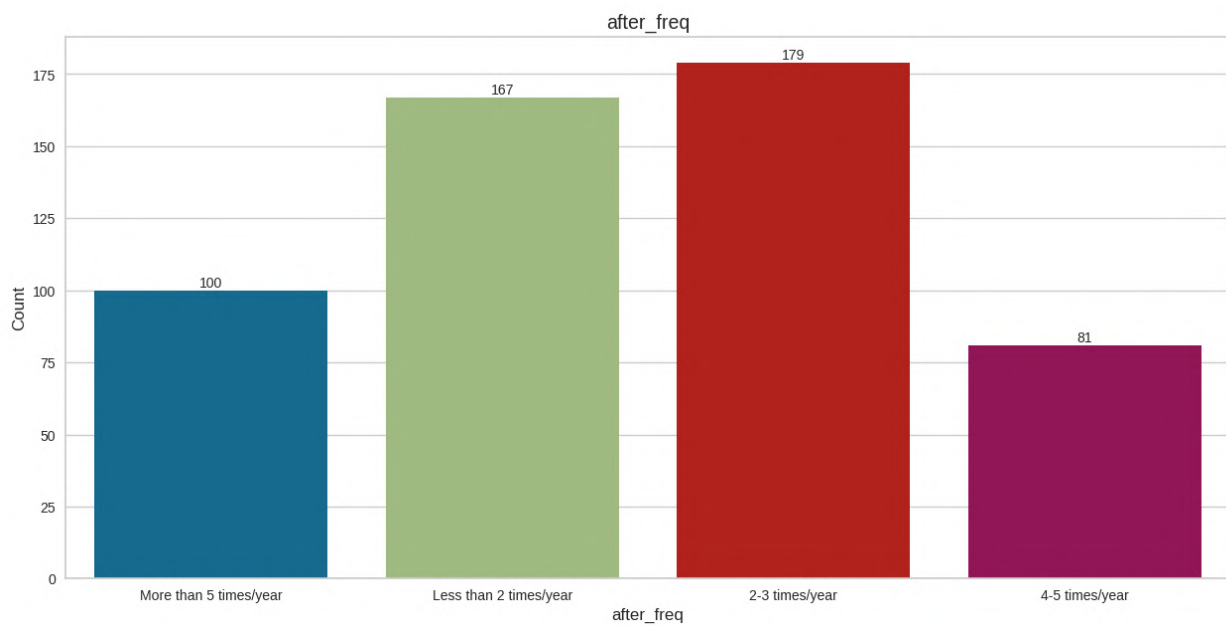


Figure 3.13 After_freq

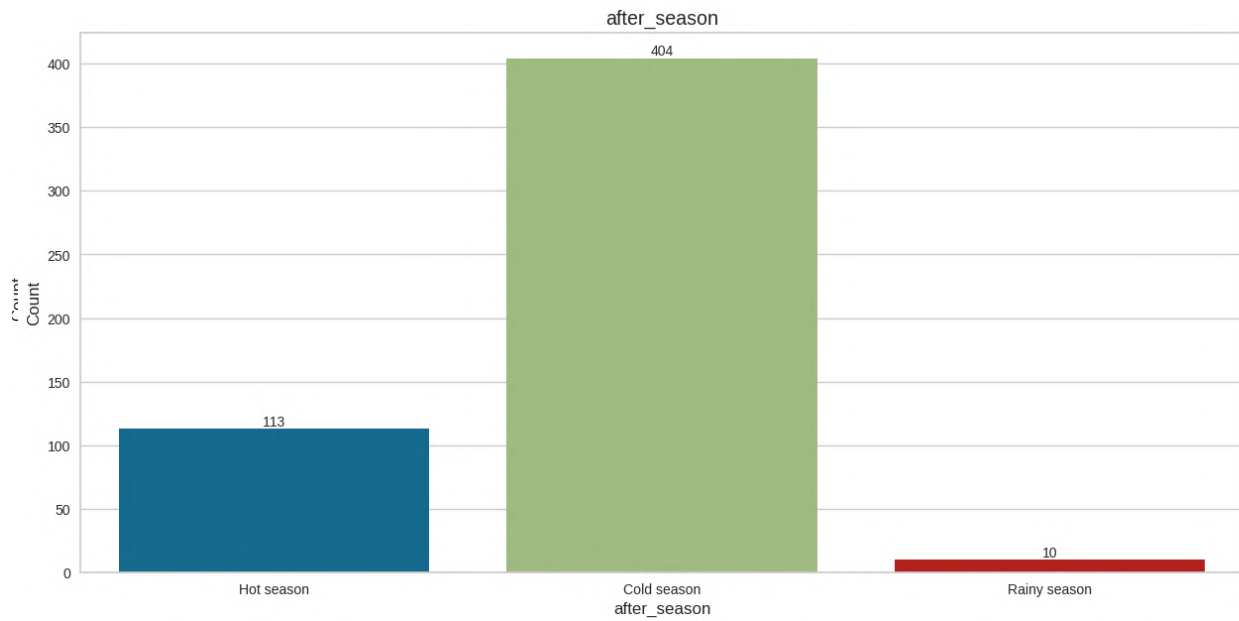


Figure 3.14 After_season

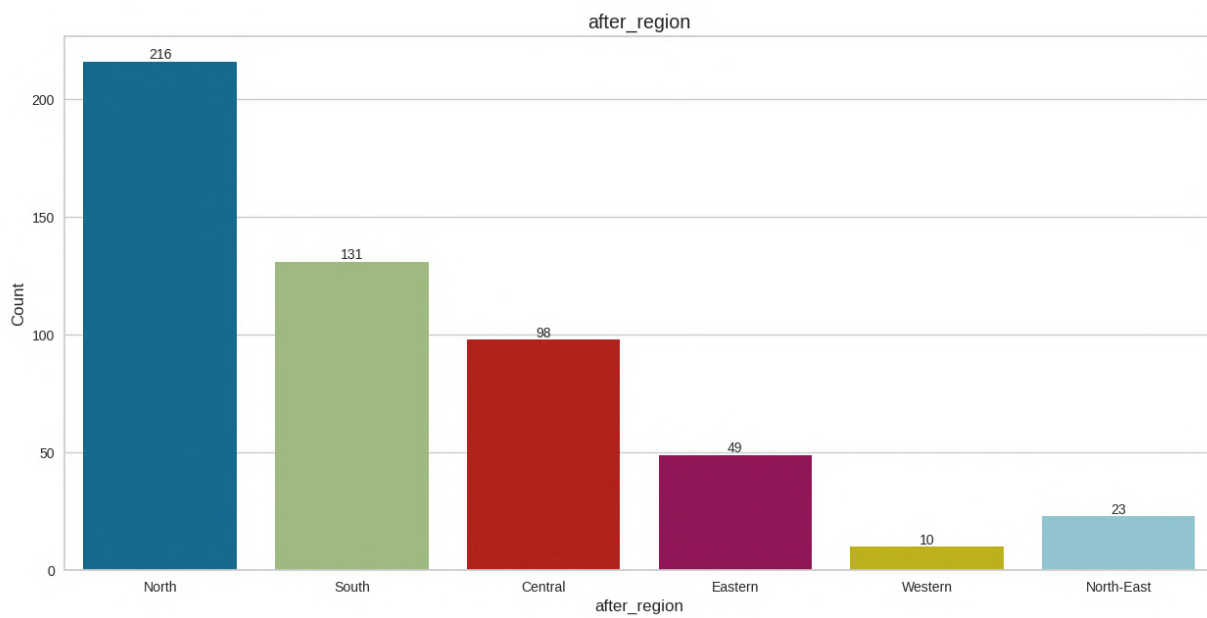


Figure 3.15 After_region

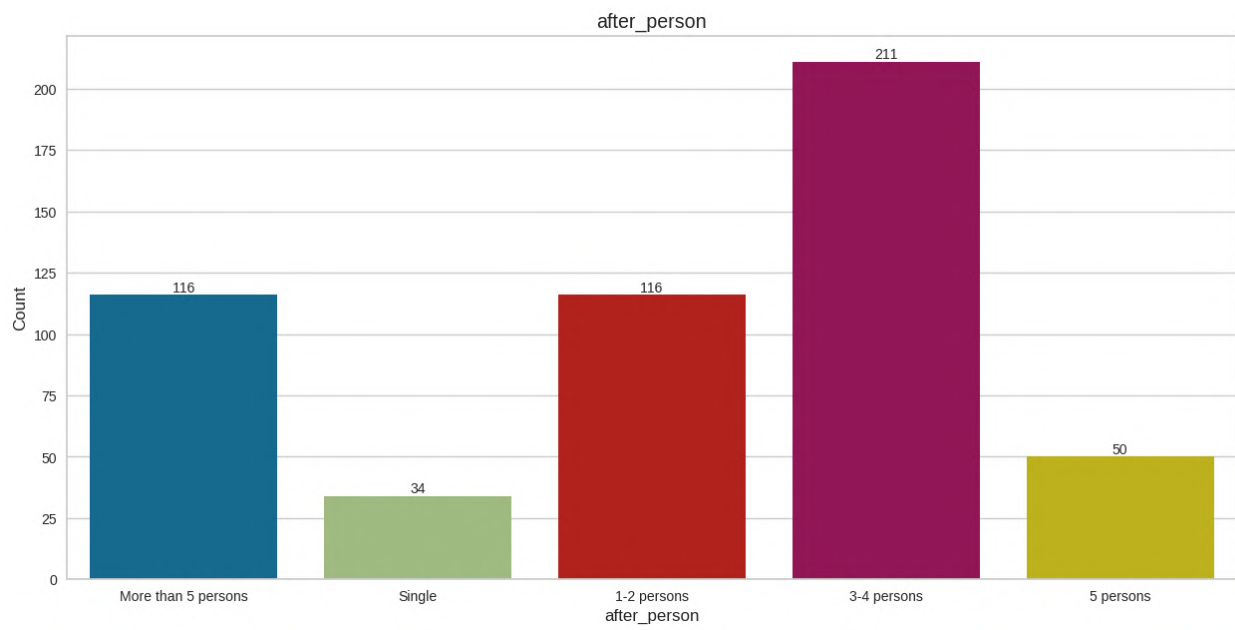


Figure 3.16 After_person

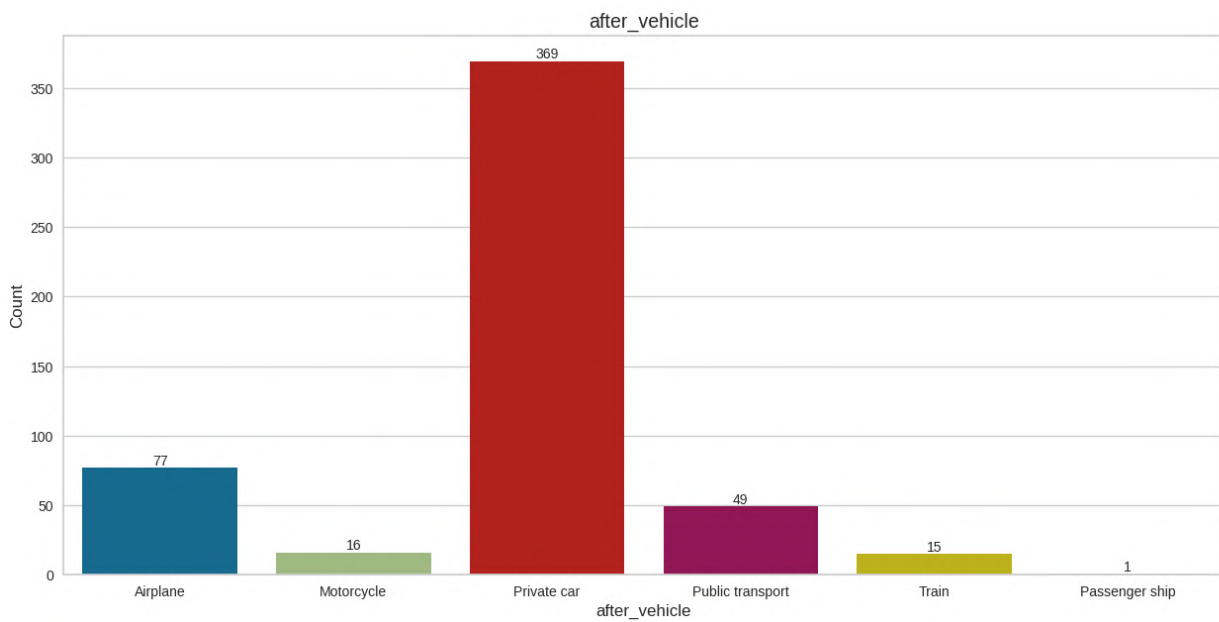


Figure 3.17 After_vehicle

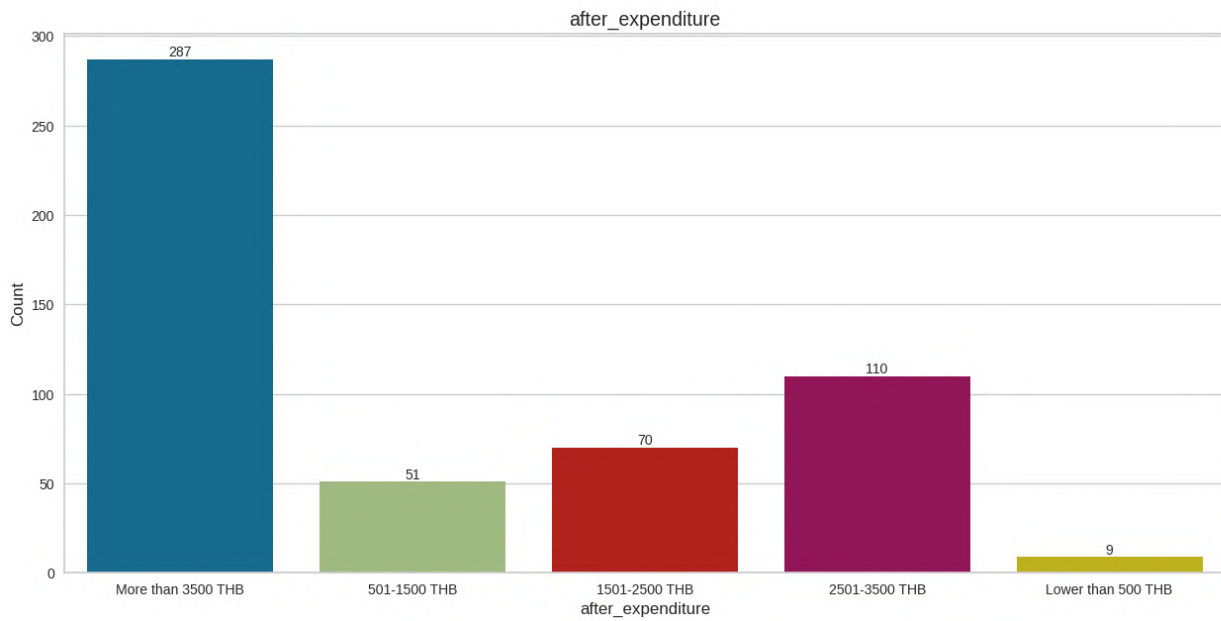


Figure 3.18 After_expenditure

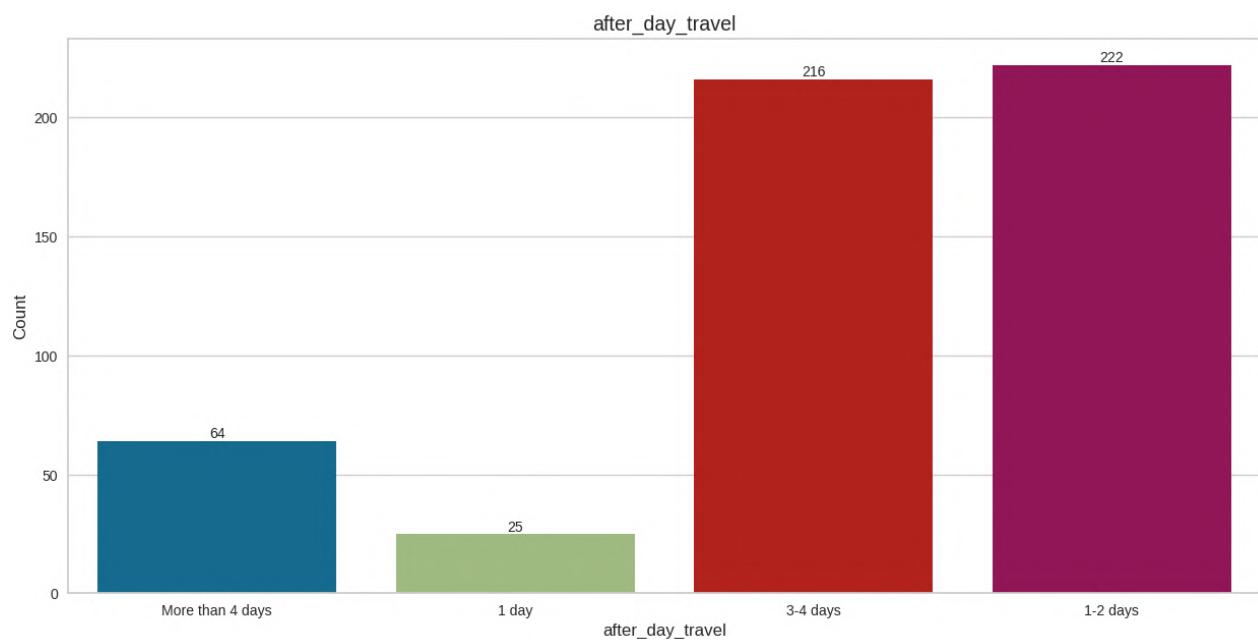


Figure 3.19 After_day_travel

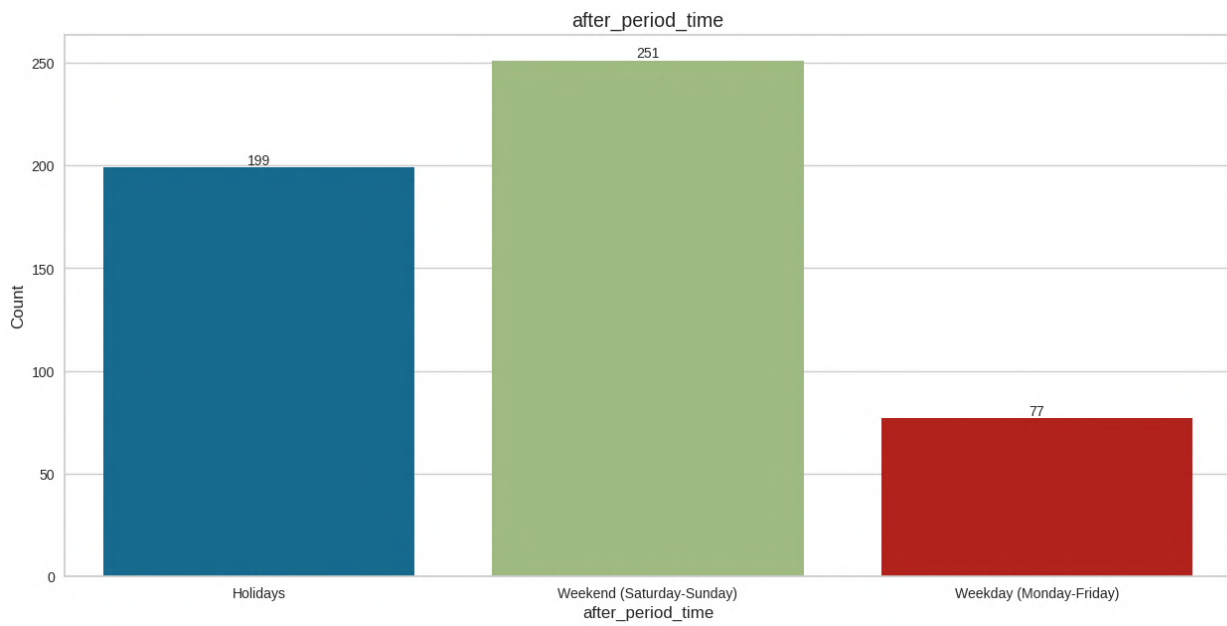


Figure 3.20 After_period_time

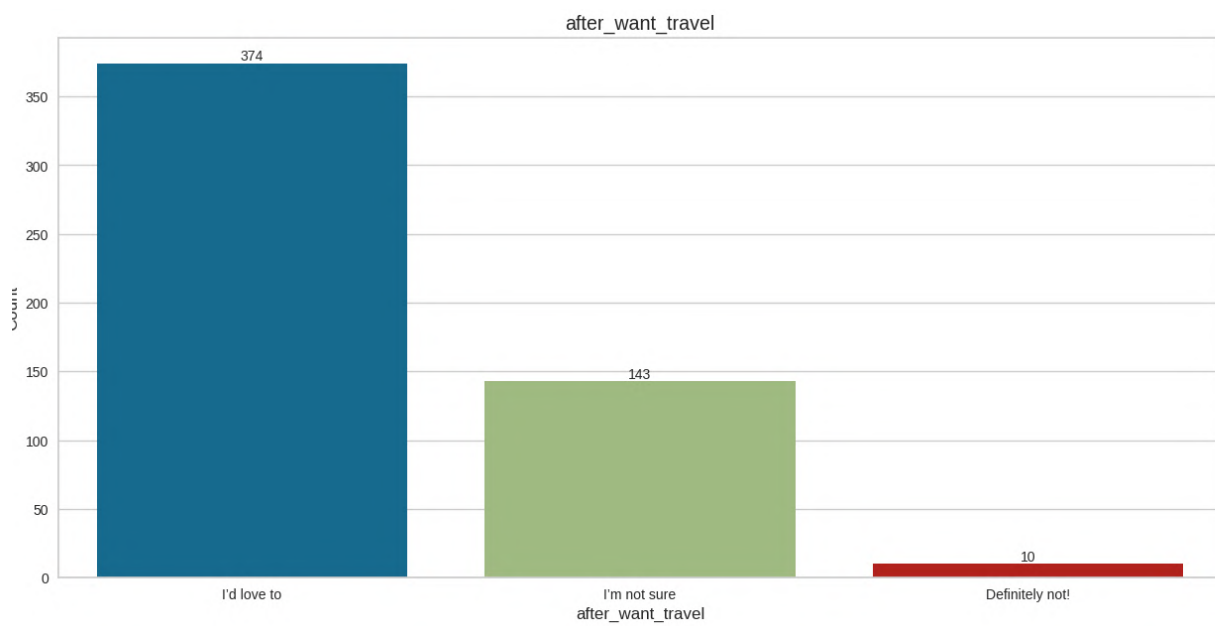


Figure 3.21 After_want_travel

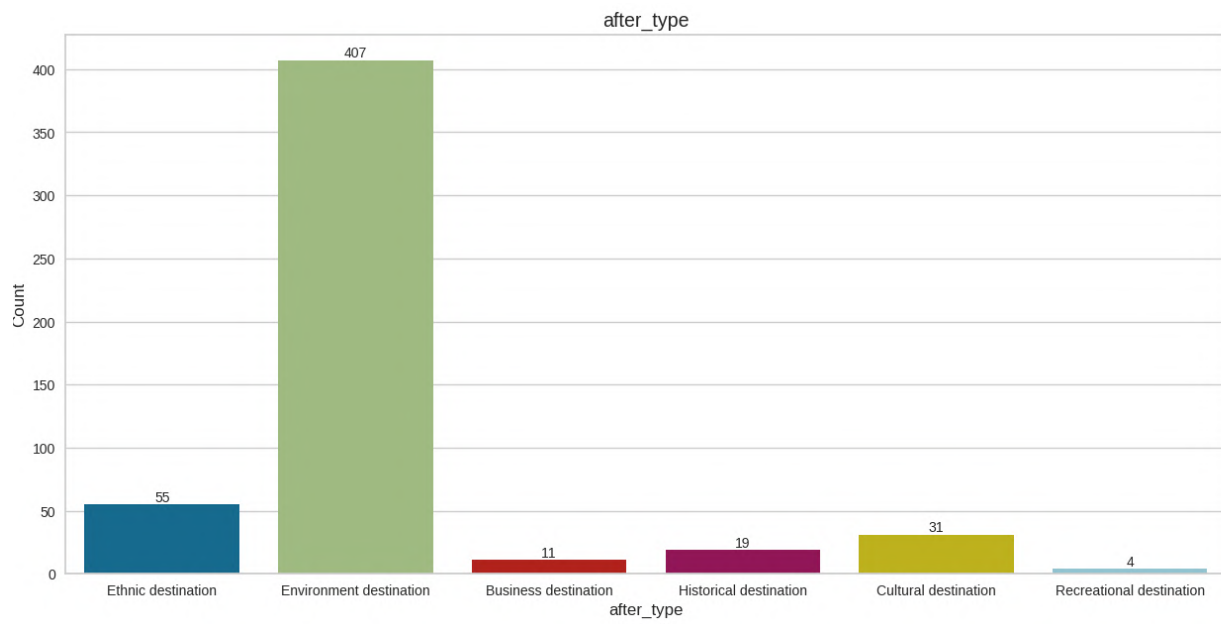


Figure 3.22 After_type

3.3 Data pre-processing

3.3.1 Check missing value

Showing the missing value of each feature.

```

age                                0
education                          0
income                             0
household                          0
after_freq                         0
after_person                       0
after_expenditure                  0
after_day_travel                   0
gender                             0
occupation                        0
per_vehicle                        0
status                             0
per_region                        0
after_season                       0
after_type                         0
after_region                       0
after_vehicle                      0
after_period_time                  0
after_want_travel                  0
dtype: int64

```

Figure 3.23 Shows missing data

3.3.2 Replace data

To replace value data that have the same value for comfortable during encode data process

```

replace_gender = {'ชาย': 'เพศชาย', 'หญิง': 'เพศหญิง'}
df['gender'] = df['gender'].replace(replace_gender)

```

```

replace_occupation = {'นักเรียน, นักศึกษา': 'นักเรียน,นักศึกษา'}
df['occupation'] = df['occupation'].replace(replace_occupation)

```

```

replace_household = {'1คน': '1 คน'}
df['household'] = df['household'].replace(replace_household)

```

```

replace_region = {'ภาคตะวันออกเฉียงเหนือ(ภาคอีสาน)': 'ภาคตะวันออกเฉียงเหนือ'}
df['per_region'] = df['per_region'].replace(replace_region)

```

```

replace_after_type = {'แหล่งท่องเที่ยวเพื่อกิจกรรมทางธุรกิจ(เช่น สัมมนา, เจรจาทริค, ประชุม)': 'แหล่งท่องเที่ยวเพื่อธุรกิจ ( เช่น การสัมมนา,เจรจาธุรกิจ,ประชุม)',
                      'แหล่งท่องเที่ยวเพื่อศึกษากลุ่มชาติพันธุ์(เช่น หมู่บ้านกะเหรี่ยง, workshopปลูกกาแฟ, home stay)': 'แหล่งท่องเที่ยวเพื่อศึกษากลุ่มชาติพันธุ์ ( เช่น หมู่บ้าน',
                      'แหล่งท่องเที่ยวเพื่อศึกษากลุ่มชาติพันธุ์ (เช่น หมู่บ้านกะเหรี่ยง, workshopปลูกกาแฟ, home stay)': 'แหล่งท่องเที่ยวเพื่อศึกษากลุ่มชาติพันธุ์ ( เช่น หมู่บ้าน'
data_a['after_type'] = data_a['after_type'].replace(replace_after_type)

replace_after_region = {'ภาคตะวันออกเฉียงเหนือ(ภาคอีสาน)': 'ภาคตะวันออกเฉียงเหนือ'}
data_a['after_region'] = data_a['after_region'].replace(replace_after_region)

replace_after_expend = {'ต่ำกว่า 500 บาท': 'น้อยกว่า 500 บาท', '501-1,500 บาท': '501-1500 บาท', '1,501-2,500 บาท': '1501-2500 บาท', '2,501-3,500 บาท': '2501-3500 บาท'}
data_a['after_expenditure'] = data_a['after_expenditure'].replace(replace_after_expend)

```

Figure 3.24 Shows replace data

3.3.3 Encode data

Convert the data from categorical to numerical data by using ordinal encoder and label encoder from sklearn library.

	age	education	income	household	after_freq	after_person	after_expenditure	after_day_travel	gender	occupation	per_vehicle	status	per_region
0	1.0	1.0	0.0	2.0	3.0	4.0	4.0	3.0	2	2	1	1	5
1	1.0	1.0	0.0	2.0	0.0	0.0	1.0	0.0	0	2	1	1	4
2	1.0	1.0	0.0	2.0	1.0	4.0	4.0	2.0	0	2	2	1	4
3	1.0	1.0	0.0	2.0	0.0	1.0	4.0	1.0	2	2	2	1	5
4	1.0	1.0	0.0	2.0	2.0	1.0	2.0	2.0	2	2	0	1	5
...	...	...	...	...	...	...	...	...	...	...	...	...	...
522	3.0	1.0	1.0	1.0	1.0	4.0	3.0	2.0	2	3	3	2	0
523	1.0	1.0	0.0	1.0	1.0	1.0	4.0	1.0	2	3	3	1	5
524	2.0	1.0	1.0	3.0	0.0	4.0	4.0	1.0	2	3	3	1	0
525	1.0	0.0	0.0	1.0	1.0	2.0	4.0	1.0	2	2	3	1	0
526	2.0	1.0	2.0	1.0	0.0	2.0	4.0	1.0	2	3	0	1	0

527 rows × 19 columns

Figure 3.25 Shows the data that was converted from categorical to numerical

3.3.4 Standardization

Standardized the data to handle outliers by using standardscaler(z-score).

	age	education	income	household	after_freq	after_person	after_expenditure	after_day_travel	gender	occupation	per_vehicle	status
0	-0.562200	0.363951	-0.800530	0.435154	1.640062	1.522884	0.763519	1.836593	0.572621	-0.446389	-0.374552	0.234921
1	-0.562200	0.363951	-0.800530	0.435154	-1.118382	-1.835107	-1.986192	-2.113956	-1.822388	-0.446389	-0.374552	0.234921
2	-0.562200	0.363951	-0.800530	0.435154	-0.198901	1.522884	0.763519	0.519743	-1.822388	-0.446389	0.451343	0.234921
3	-0.562200	0.363951	-0.800530	0.435154	-1.118382	-0.995609	0.763519	-0.797107	0.572621	-0.446389	0.451343	0.234921
4	-0.562200	0.363951	-0.800530	0.435154	0.720580	-0.995609	-1.069622	0.519743	0.572621	-0.446389	-1.200447	0.234921
...	...	...	...	...	...	...	...	...	...	...	...	...
522	1.201368	0.363951	0.185169	-0.875281	-0.198901	1.522884	-0.153052	0.519743	0.572621	0.227671	1.277238	2.264488
523	-0.562200	0.363951	-0.800530	-0.875281	-0.198901	-0.995609	0.763519	-0.797107	0.572621	0.227671	1.277238	0.234921
524	0.319584	0.363951	0.185169	1.745588	-1.118382	1.522884	0.763519	-0.797107	0.572621	0.227671	1.277238	0.234921
525	-0.562200	-1.573444	-0.800530	-0.875281	-0.198901	-0.156112	0.763519	-0.797107	0.572621	-0.446389	1.277238	0.234921
526	0.319584	0.363951	1.170869	-0.875281	-1.118382	-0.156112	0.763519	-0.797107	0.572621	0.227671	-1.200447	0.234921

527 rows × 19 columns

Figure 3.26 Shows the data that was standardized

3.4 Evaluation metric

3.4.1 Silhouette coefficient

The Silhouette coefficient evaluates the cluster based on the measure of the average distance between one data point and other data points in the same clusters and the average distance among different clusters.

3.4.2 ELbow Method

A heuristic used in determining the number of clusters in a data set. The method consists of plotting the explained variation as a function of the number of clusters and picking the elbow of the curve as the number of clusters to use.

3.5 Model

3.5.1 K-means clustering

The method of vector quantization, originally from signal processing, aims to partition n observations into k clusters in which each observation belongs to the cluster with the nearest mean (cluster centers or cluster centroid), serving as a prototype of the cluster.

3.5.2 Hierarchical clustering

The cluster is based on the principle that every object is connected to its neighbors depending on their proximity distance (degree of relationship). The clusters are represented in extensive hierarchical structures separated by a maximum distance required to connect the cluster parts.

3.6 Web page design and database workflow:

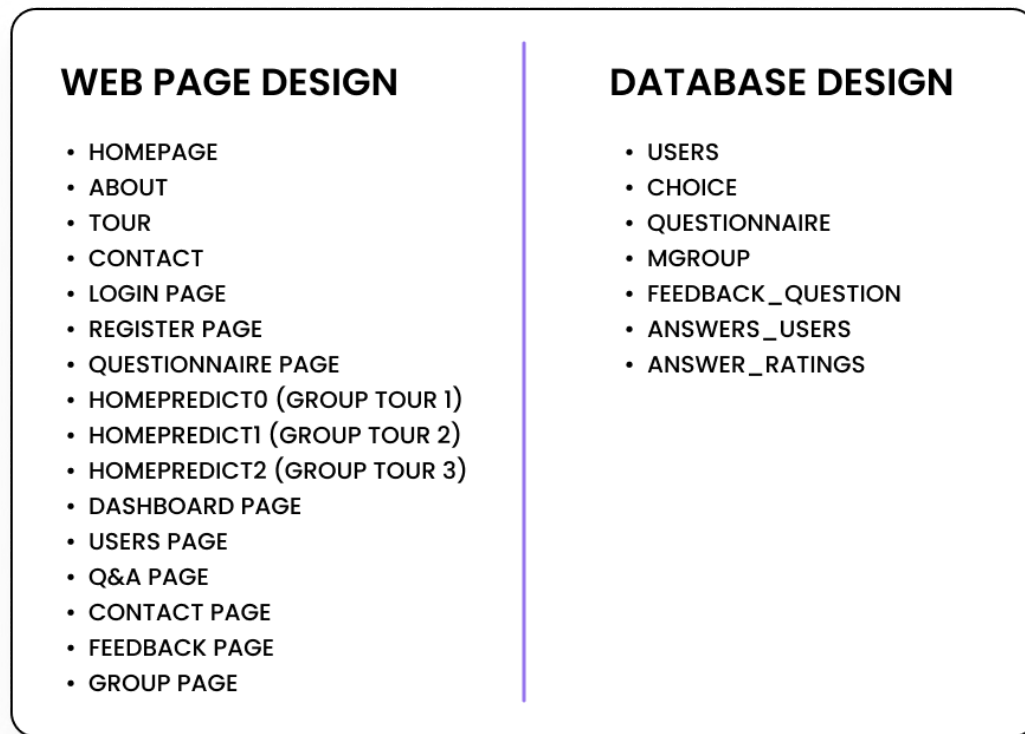


Figure 3.27 Web and database workflow

3.6.1 Web page view before-after login of user and admin

		Homepage	About	Tours	Contact	Login Page	Register Page	Questionnaire Page	HomePredict0 (Group Tour 1)	HomePredict1 (Group Tour 2)	HomePredict2 (Group Tour 3)	Dashboard Page	Users Page	Q&A Page	Feedback Page	Group Page
User	Before login	✓	✓		✓	✓	✓									
	After login	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓					
Admin	View	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Edit												✓			✓
	Delete												✓			

Figure 3.28 Show which page the admin and user can view

3.6.2 Web page design

1. Register/Login

The image displays two web page designs for 'Rocket Trips'. The top design is the 'Sign up' page, featuring an orange background with large, overlapping circles. It includes a title 'Start your adventure with Rocket Trips', a sub-header 'Create an account to embark on a seamless travel experience...', and a registration form with fields for Name, Email, Password, and Confirm Password, followed by a 'REGISTER' button and a 'Go to Login' link. The bottom design is the 'Log In' page, featuring a dark blue background with large, overlapping circles. It includes a title 'Welcome back Continue your journey', a sub-header 'Sign in to access your personalized travel itineraries...', and a login form with fields for Email and Password, followed by a 'LOG IN' button and a 'Register here' link.

Sign up

Name

Email

Password

Confirm Password

REGISTER

[Go to Login](#)

Log In

Please enter your username and password

Email

Password

LOG IN

[I don't have an account ? Register here](#)

Figure 3.29 Sign up/Login page

- The registration process necessitates the provision of specific details, including your name, email, password, and confirmation of the password. It is imperative that the user password adheres to the following criteria: it must be a minimum of 8 characters in length, encompass at least one uppercase letter, integrate at least one numeric character, and include at least one special character.
- In case of login failure due to an incorrect username or password, the system will generate an alert message to notify the user.

2. Home page

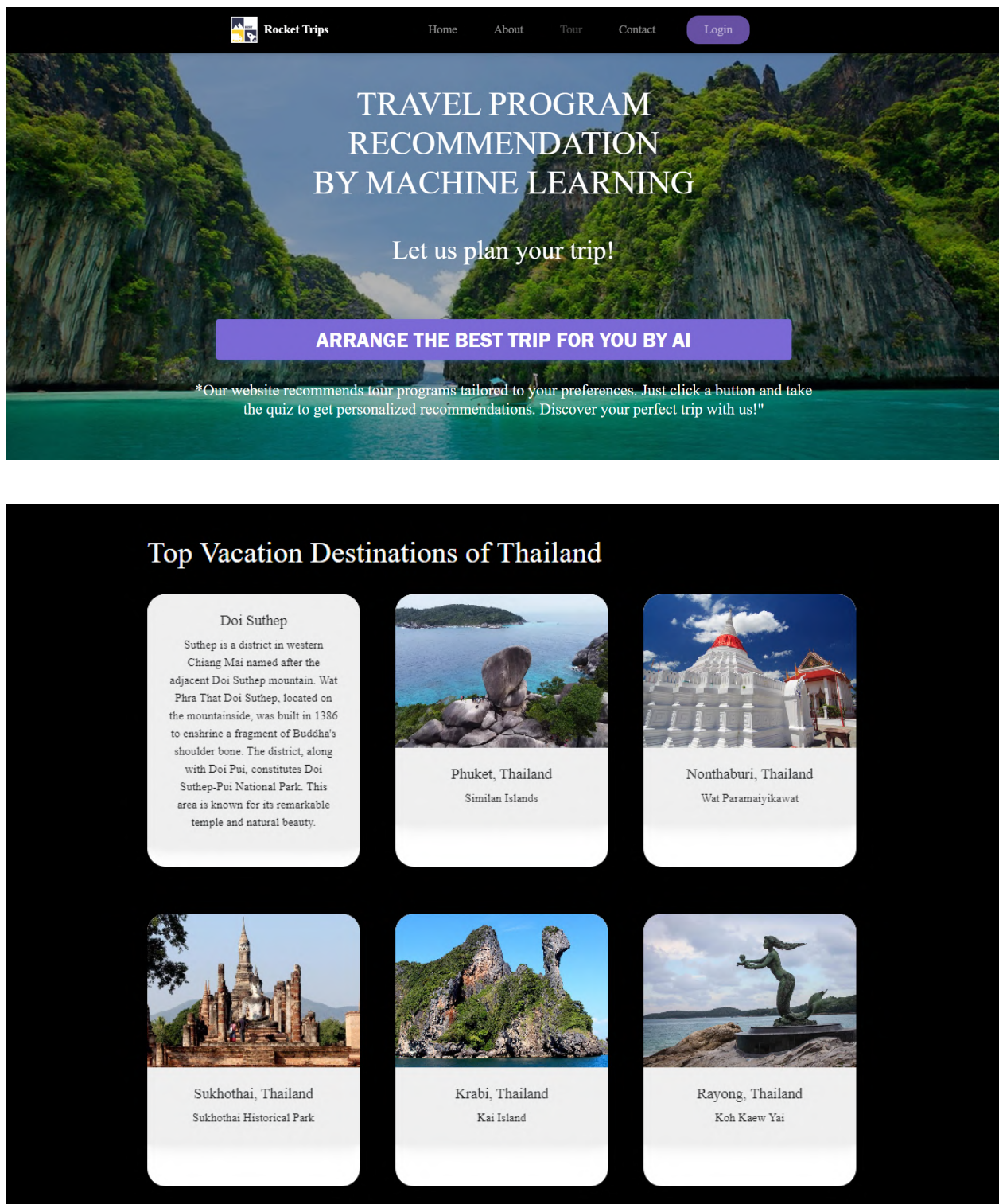


Figure 3.30 Home page

- Clicking on the "Arrange the best trip for you by AI" button, you will be directed to a questionnaire designed to gather information for the purpose of making predictions to provide you with the best possible travel itinerary.
- Whenever a user hovers their cursor over a card element, the card will flip to reveal detailed information and a narrative about the recommended location.

3. About page



Rocket Trips

Rocket Trips provides a range of travel experiences, from exciting adventures to relaxing getaways, designed to enhance your journeys. Our machine learning tools will help you find the perfect trip based on your preferences and aspirations. Using machine learning clustering, we've organized our packages into four subgroups, each with unique features and attributes. Embark on a journey that elevates your travel experience with Rocket Trips.

Clustering Insights: Categorized into 3 Main Groups



Characteristics of Group1

This group consists of female and male mostly graduated under bachelor's degree with age range less than 20 – 40 years old which have an income less than 15,000 – 45,000 baht per month , most of the member in this group have their own motor cycle , their home town mostly located in Western , North-East and Central of Thailand , The majority of them status is single with 1-3 people in their households , their preferences for traveling is 1-2 days per trip , mostly traveling with 1-4 people and they prefer to travel at the Central , Eastern , North-East and Western of Thailand , They like to travel during hot and cold season and most like to travel in cultural including historical and recreational , with a budget of less than 500 – 2500 baht per person.



Characteristics of Group2

This group consists of a male and mostly a working age which more than 31 years old, graduated higher than bachelor's degree level. Their income is mostly more than 30,000 baht per month, mostly people in this group have their own private car, the majority of them status are married and they prefer to travel in historical including ethnic with a budget of more than 3500 baht per person.



Characteristics of Group3

This group is consisting of Alternative and female, most of them are students graduated Bachelor's degree or studying, with age less than 20 – 30 years old, Their income mostly less than 15,000 baht per month, They mostly own the other vehicle apart from the private car and also own a motorcycle and some of them don't have any of it. Their hometown located in eastern, south, north-east and the north of Thailand, the majority of their status is single with more than 4 peoples in their household, They prefer to travel for 3-4 or more than 4 days per trip with more than 5 peoples in each trip. They mostly like to travel with airplane or private car, they like to travel during both cold and hot season and like to travel in recreational and ethnic with a budget more than 2,500 baht per person. They mostly travel during weekends or holidays and lovely to travel more than twice per years

Figure 3.31 About page

- Provides an overview of the website's purpose and functionality.
- Describing the characteristics of each group derived from machine learning predictions.

5. Tour Page

- 5.1 Tour Page Group 1

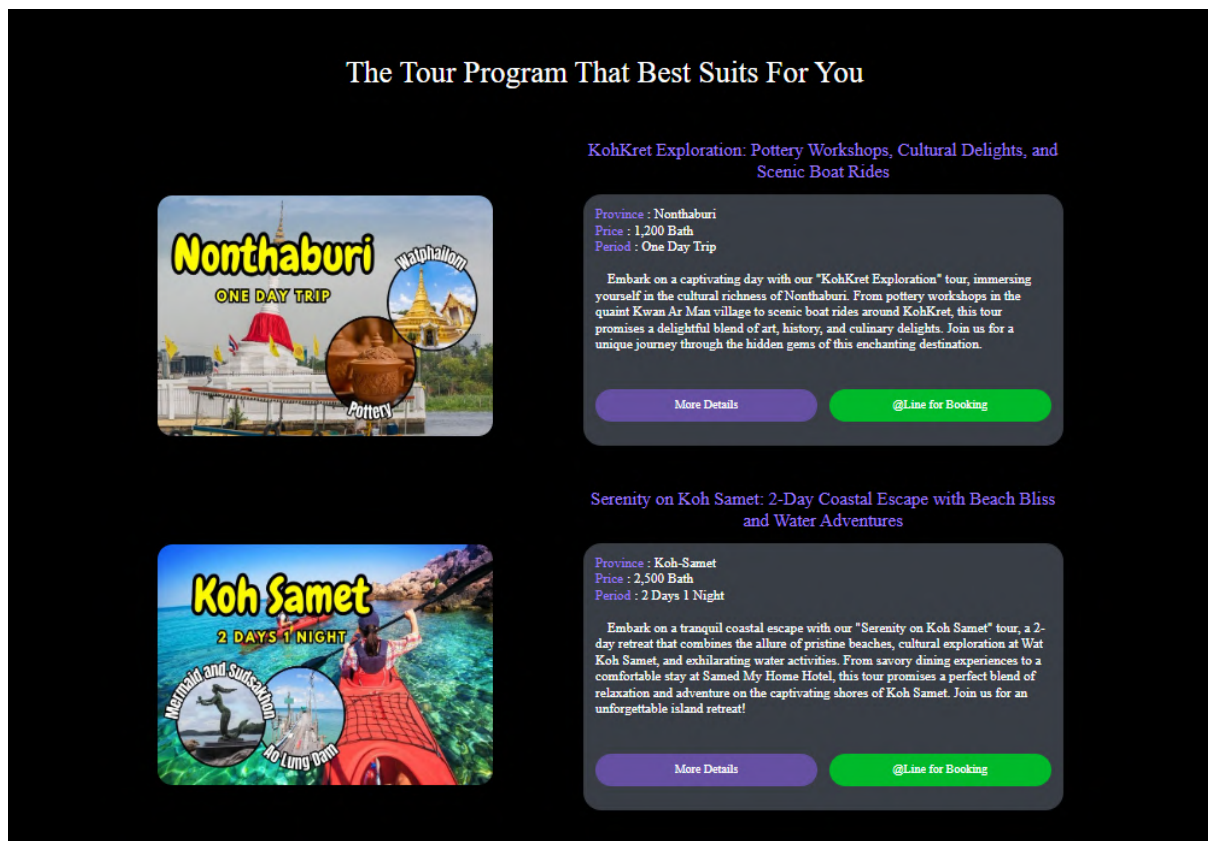
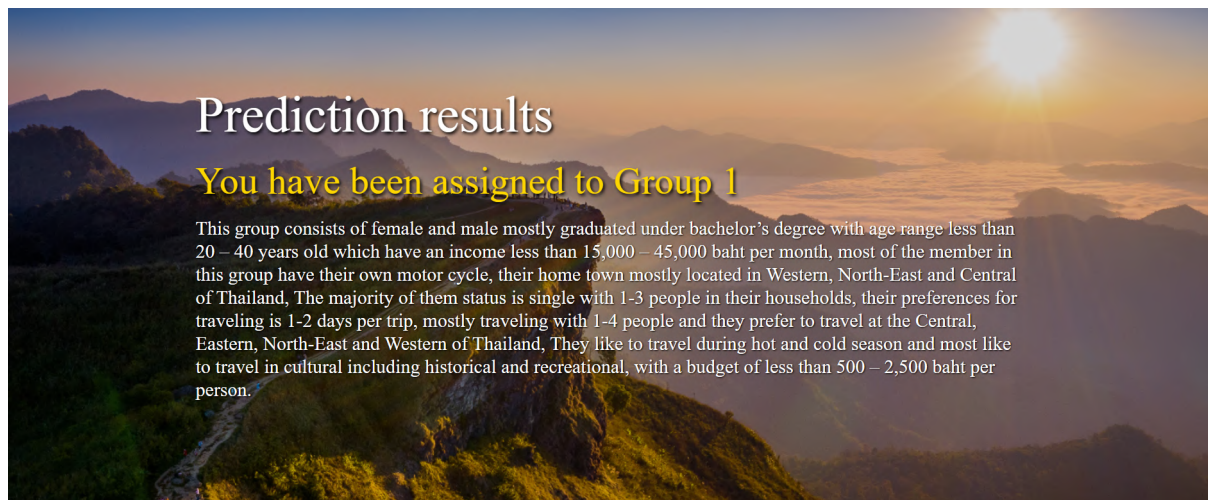


Figure 3.32 Tour page group 1

You have been assigned to Group 2

This group consists of a male and mostly a working age which more than 31 years old, graduated higher than bachelor's degree level. Their income is mostly more than 30,000 baht per month, mostly people in this group have their own private car, the majority of them status are married and they prefer to travel in historical including ethnic with a budget of more than 3500 baht per person.

Immerse Yourself in Krabi: A 2-Day Journey Through Local Heritage and Scenic Marvels



Province : Krabi
Price : 3,100 Bath
Period : 2 Days 1 Nights

Embark on an enriching 2-day adventure with our "Discover Krabi's Charm" tour. Immerse yourself in the cultural tapestry of Lanta Old Town, savor delectable meals at NAHM restaurant, and explore historical landmarks at Wat Maha Thachiramongkol. From the vibrant markets of OTOP to the pristine beaches of Promthep Cape, this tour seamlessly blends cultural exploration, culinary delights, and breathtaking natural wonders. Join us for an unforgettable journey through the captivating landscapes of Krabi!

[More Details](#)

@Line for Booking

Explore Sukhothai: Embark on a 3-Day Journey through Ancient Temples, Artistic Workshops, and Local Heritage.



Province : Sukhothai
Price : 3,800 Bath
Period : 3 Days 2 Night

Embark on an enriching 3-day adventure with our "Explore Sukhothai" tour. Immerse yourself in the ancient charm of temples, engage in artistic workshops, and delve into the local heritage of Sukhothai. This carefully curated itinerary promises a unique blend of cultural exploration and hands-on experiences, providing an unforgettable journey through the heart of Thailand's historical treasures. Join us for an immersive and insightful exploration!

[More Details](#)

@Line for Booking

Figure 3.33 Tour page group 2

- 5.3 Tour Page Group 3

Prediction results

You have been assigned to Group 3

This group is consisting of Alternative and female, most of them are students graduated Bachelor's degree or studying , with age less than 20 – 30 years old , Their income mostly less than 15,000 baht per month , They mostly own the other vehicle apart from the private car and also own a motorcycle and some of them don't have any of it . Their hometown located in eastern , south , north-east and the north of Thailand , the majority of their status is single with more than 4 peoples in their household and they prefer to travel at the Western, North and South of Thailand , They prefer to travel for 3-4 or more than 4 days per trip with more than 5 peoples in each trip. They mostly like to travel with airplane or private car , they like to travel during both cold and hot season and like to travel in recreational and ethnic with a budget more than 2,500 baht per person. They mostly travel during weekends or holidays and lovely to travel more than twice per years.

The Tour Program That Best Suits For You



Majestic Chiang Mai Adventure: 3-Day Retreat with Elephant Encounters, Scenic Landscapes, and Cultural Discoveries.

Province : Chiang Mai
Price : 4,600 Bath
Period : 3 Days 2 Night

Embark on an unforgettable 3-day adventure with our "Majestic Chiang Mai" tour. Immerse yourself in the rich culture at Wat Phra That Doi Suthep, experience the thrill of Jungle Coaster Pongyang, and interact with gentle giants at Maesa Elephant Camp. This meticulously crafted itinerary combines cultural exploration, thrilling activities, and natural wonders, offering a truly immersive journey through the heart of Chiang Mai. Join us for an extraordinary exploration of northern Thailand's hidden gems!

[More Details](#) [@Line for Booking](#)



Rapids, Waterfalls, and Border Markets Adventure

Province : Phuket
Price : 3,900 Bath
Period : 3 Days 2 Night

Embark on an exhilarating journey and dive into the heart of nature as you conquer challenging rapids, explore vibrant villages, and witness the beauty of waterfalls. Camp under the stars, engage in cultural encounters, and discover hidden gems along the Burmese border markets. This tour promises an immersive experience, blending adventure, cultural discovery, and the tranquility of the wilderness.

[More Details](#) [@Line for Booking](#)

Figure 3.34 Tour page group 3

- The tour program will be recommended based on predictions from the user's questionnaire, utilizing machine learning algorithms and dividing the group into 3 groups.
- If users wish to learn more about our tour packages, they can click the 'More Details' button to access additional information. For those interested in

purchasing tour packages, simply click the '@Line for Booking' button on the website page, which will lead users to the Contact page.

Find perfect TRIP with us!

Please take a few minutes to complete our survey and provide us with your valuable feedback

Arrange the best trip for you by AI

We need your feedback

We utilize Machine Learning to suggest tourist attractions tailored to your interests. We'd appreciate your feedback on how well our recommendations align or don't align with your preferences. This will help us enhance our system, leading to more precise travel suggestions in the future.

Question 1: According to the prediction results from machine learning, how closely do the characteristics match yours?

1 2 3 4 5

Question 2: How well did our recommended tour package align with your personal interests and preferences?

1 2 3 4 5

Question 3: How helpful were the tour recommendations in assisting you to find a suitable tour package?

1 2 3 4 5

Question 4: How worthwhile and interesting are the tour packages we recommend?

1 2 3 4 5

Question 5: How appropriate are the questions in our questionnaire?

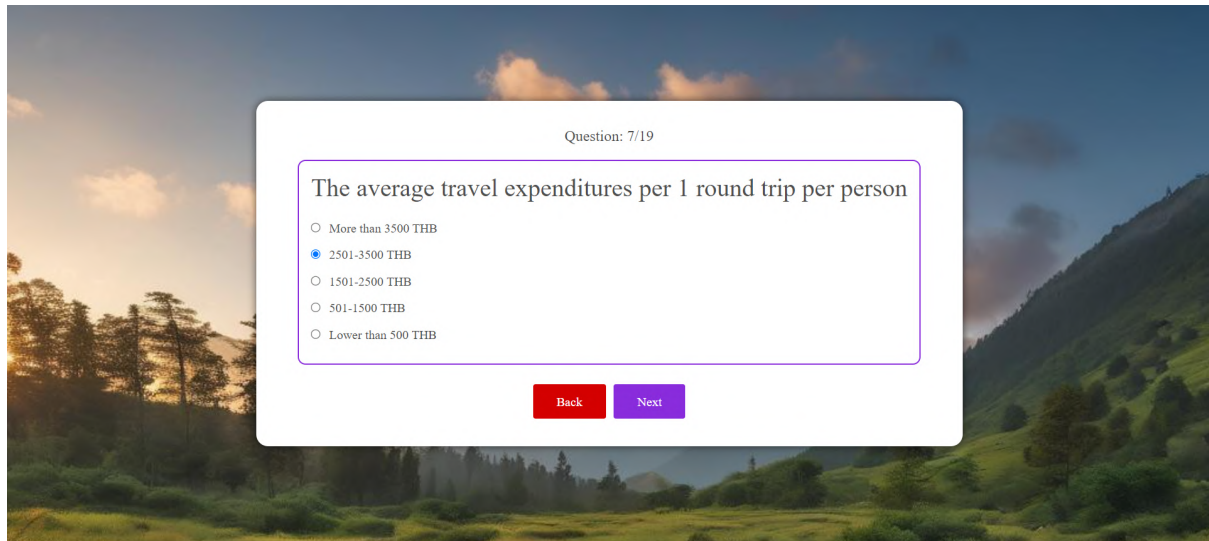
1 2 3 4 5

Reset Submit

Figure 3.35 Tour page

- If the predicted results do not meet your expectations, you can click the "Arrange the best trip for you by AI" button to fill out another questionnaire and receive prediction results that better align with your preferences.
- Users have the ability to express their level of contentment or discontentment with each package by assigning a rating ranging from 1 to 5.

4. Questionnaire for prediction



Question: 7/19

The average travel expenditures per 1 round trip per person

☐ More than 3500 THB

☒ 2501-3500 THB

☐ 1501-2500 THB

☐ 501-1500 THB

☐ Lower than 500 THB

Back Next

Figure 3.36 Questionnaire page

- The questionnaire page enables users to respond to a set of defined questions, and upon completion, the information provided will be analyzed and predicted in the subsequent step.

5. Contact/Footer

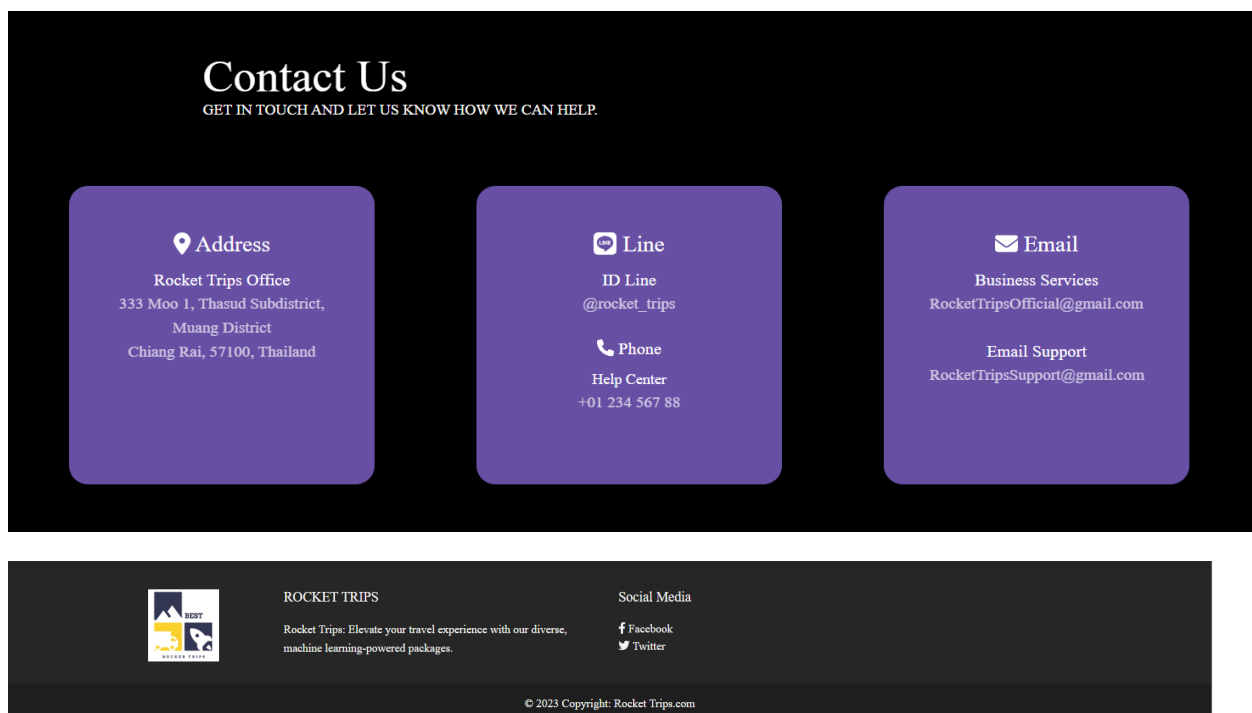


Figure 3.37 Contact/Footer

6. Admin

6.1 Dashboard

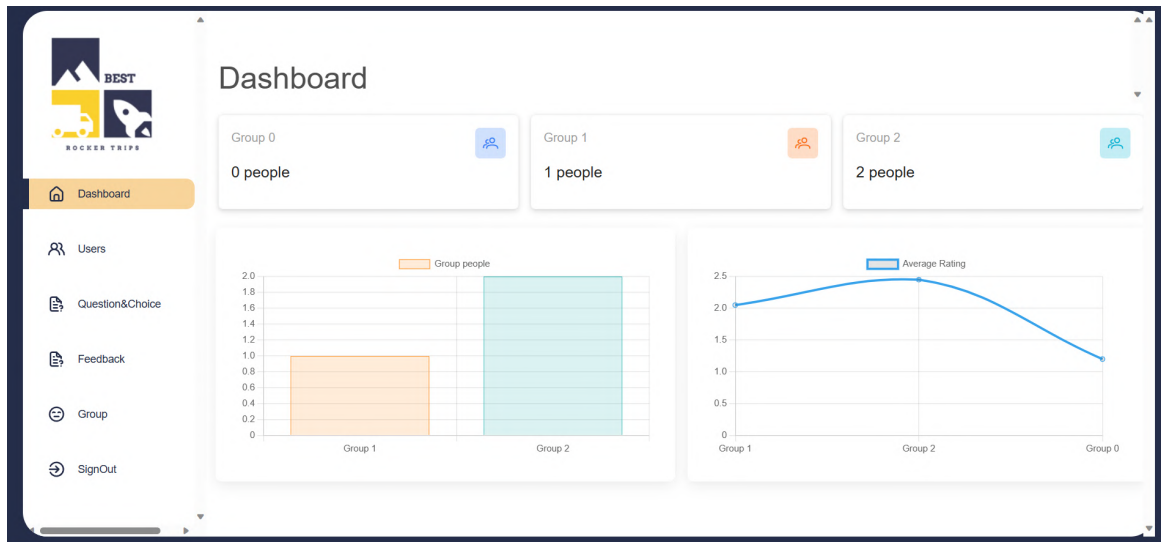


Figure 3.38 Dashboard

- The dashboard prominently presents the total number of individuals in each group through the use of top cards. Additionally, it features a bar graph for a comparative view of the population in each group. A line graph is also integrated, showcasing the scoring metrics for each group.

6.2 User page

The Users Table displays the following data:

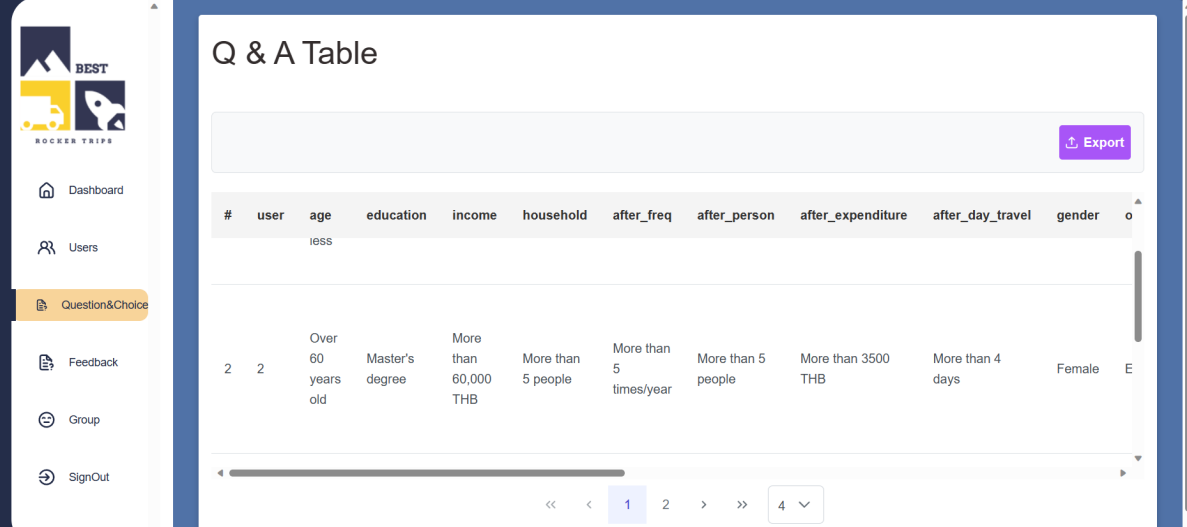
#	Username ↑↓	Email	Role	Group	Feedback	
1	ton	ton@gmail.com	USER	1	1	Edit Delete
2	pop	pop@gmail.com	USER	2	1	Edit Delete
3	ply	ply@gmail.com	USER	2	1	Edit Delete

The table includes an 'Export' button and a pagination control showing 1 of 4 items.

Figure 3.39 User table

- This page includes a comprehensive table detailing all user information. It highlights the group affiliation of each user. Furthermore, the distribution of users across different groups is visualized through a bar graph. Administrators have the capability to modify user roles and remove any unwanted user accounts.

6.3 Question and choice



Q & A Table

Export

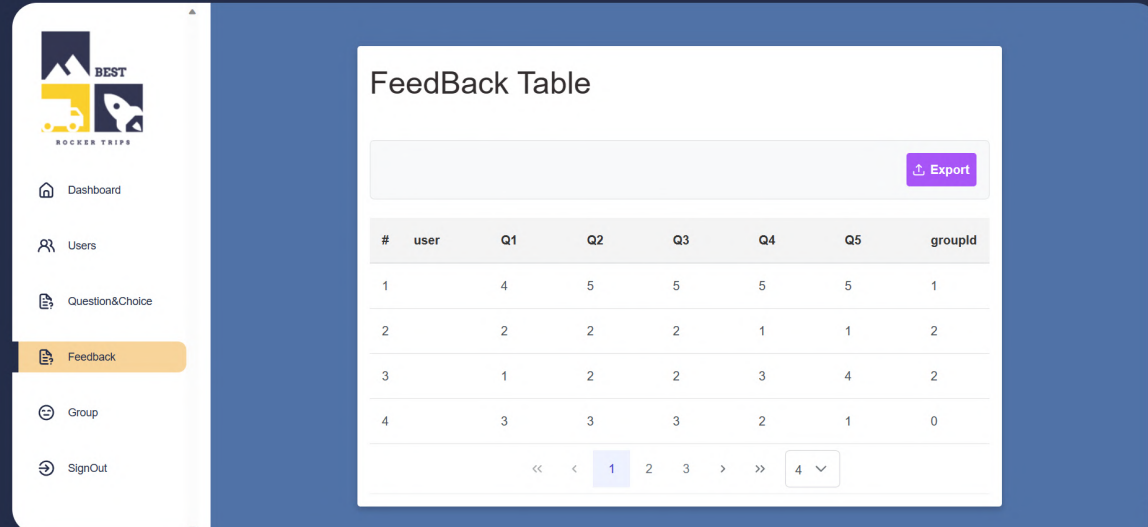
#	user	age	education	income	household	after_freq	after_person	after_expenditure	after_day_travel	gender	o
2	2	Over 60 years old	Master's degree	More than 60,000 THB	More than 5 people	More than 5 times/year	More than 5 people	More than 3500 THB	More than 4 days	Female	E

Navigation: << < 1 2 > >> 4

Figure 3.40 Question and choice table

- The Q&A section presents a structured table that captures user responses to various questions alongside the corresponding predictions made by the model.

6.4 Feedback



FeedBack Table

Export

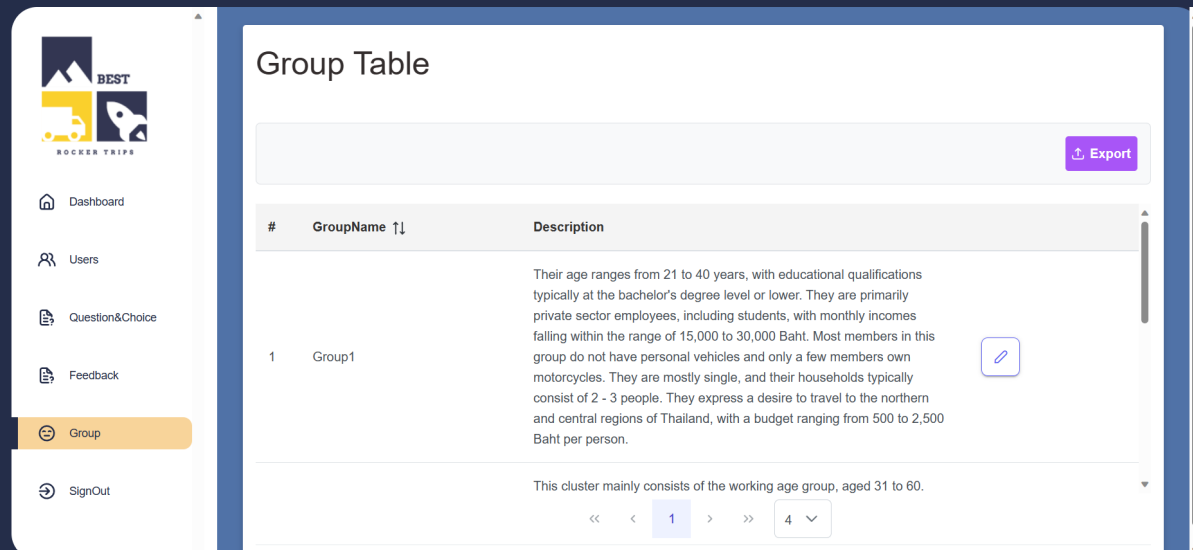
#	user	Q1	Q2	Q3	Q4	Q5	groupId
1	4	5	5	5	5	5	1
2	2	2	2	2	1	1	2
3	1	2	2	3	4	4	2
4	3	3	3	2	1	1	0

Navigation: << < 1 2 3 > >> 4

Figure 3.41 Feedback table

- In this section, a table is provided to review user feedback on each question. This feedback is quantitatively assessed, and the average results are displayed as a graph on the Dashboard for easy analysis and interpretation.

6.5 Group



#	GroupName	Description
1	Group1	Their age ranges from 21 to 40 years, with educational qualifications typically at the bachelor's degree level or lower. They are primarily private sector employees, including students, with monthly incomes falling within the range of 15,000 to 30,000 Baht. Most members in this group do not have personal vehicles and only a few members own motorcycles. They are mostly single, and their households typically consist of 2 - 3 people. They express a desire to travel to the northern and central regions of Thailand, with a budget ranging from 500 to 2,500 Baht per person. This cluster mainly consists of the working age group, aged 31 to 60.

Figure 3.42 Group table

- This feature displays a table containing detailed descriptions of each group. Administrative users have the authority to update and edit this information as required.

7. Database

ER-Diagram of database tables as can see below:

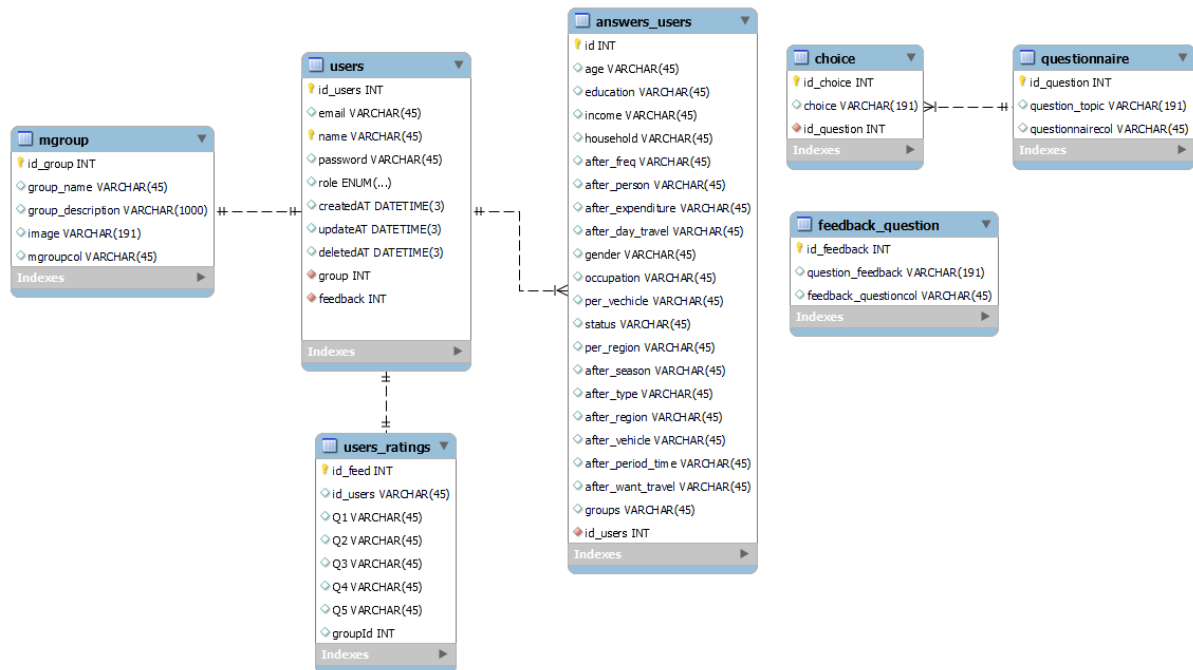


Figure 3.43 Database

8. Table

From figure 3.42 - 3.48. Show users of database table as can see below:

users

Column	Type	Null	Default	Comments
id (Primary)	int	No		
email	varchar(191)	No		
name	varchar(191)	Yes	NULL	
password	varchar(191)	No		
role	enum('ADMIN', 'USER')	No	USER	
group	int	Yes	NULL	
createdAt	datetime(3)	No	CURRENT_TIMESTAMP(3)	
updatedAt	datetime(3)	No		
deletedAt	datetime(3)	Yes	NULL	
groupId	int	Yes	NULL	
feedback	varchar(191)	Yes	NULL	

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id	0	A	No	
users_email_key	BTREE	Yes	No	email	0	A	No	

Figure 3.44 User Table

choice

Column	Type	Null	Default	Comments
id_choice (<i>Primary</i>)	int	No		
choice	varchar(191)	No		
id_question	int	No		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id_choice	0	A	No	

Figure 3.45 Choice Table

questionnaire

Column	Type	Null	Default	Comments
id_question (<i>Primary</i>)	int	No		
question_topic	varchar(191)	No		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id_question	0	A	No	

Figure 3.46 Questionnaire Table

Column	Type	Null	Default	Comments
id_feedback (<i>Primary</i>)	int	No		
question_feedback	varchar(191)	No		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id_feedback	0	A	No	

Figure 3.47 Feedback Question Table

mgroup

Column	Type	Null	Default	Comments
id_group (Primary)	int	No		
group_name	varchar(191)	No		
group_description	varchar(1000)	No		
image	varchar(191)	No		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id_group	0	A	No	

Figure 3.48 Group Table

answers_users

Column	Type	Null	Default	Comments
id (Primary)	int	No		
id_users	varchar(191)	No		
age	varchar(191)	No		
education	varchar(191)	No		
income	varchar(191)	No		
household	varchar(191)	No		
after_freq	varchar(191)	No		
after_person	varchar(191)	No		
after_expenditure	varchar(191)	No		
after_day_travel	varchar(191)	No		
gender	varchar(191)	No		
occupation	varchar(191)	No		
per_vehicle	varchar(191)	No		
status	varchar(191)	No		
per_region	varchar(191)	No		
after_season	varchar(191)	No		
after_type	varchar(191)	No		
after_region	varchar(191)	No		
after_vehicle	varchar(191)	No		
after_period_time	varchar(191)	No		
after_want_travel	varchar(191)	No		
groups	int	No		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id	0	A	No	

Figure 3.49 Answers Users Table

users_ratings

Column	Type	Null	Default	Comments
id_feed (<i>Primary</i>)	int	No		
id_users	int	No		
Q1	int	No		
Q2	int	No		
Q3	int	No		
Q4	int	No		
Q5	int	No		
groupId	int	No		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id_feed	0	A	No	

Figure 3.50 User Rating Table

CHAPTER 4

EXPERIMENT AND RESULTS

4.1 Experiment I

Compare the silhouette scores of two models on 740 samples to choose the better model for the next experiment. According to Table 4.1, K-means clustering yields a superior silhouette score, both in terms of the average and the highest score. The average silhouette score of K-means clustering is 0.118, which is 52 percent higher than hierarchical clustering. The highest score is 0.136, representing a 40 percent improvement over hierarchical clustering.

Table 4.1 shows the silhouette score between hierarchical clustering and K-means clustering.

Cluster	3	4	5	6	7	8	Average
Hierarchical	0.082	0.055	0.059	0.05	0.06	0.029	0.056
K-means	0.131	0.136	0.124	0.122	0.083	0.086	0.118

4.2 Experiment II

Compare the silhouette score of K-means model on 740 samples with PCA and without PCA to choose the better model to apply in the next experiment. From Table 4.2, the better silhouette score is K-means clustering due to the average and the highest silhouette score. The average silhouette of K-means clustering with PCA is 0.356, more than without PCA at 69 percent. The highest score is 0.136, which is more than without PCA at 40 percent.

Table 4.2 shows the silhouette score between K-means clustering with PCA and without PCA.

Cluster	3	4	5	6	7	8	Average
Without PCA	0.131	0.136	0.124	0.122	0.083	0.086	0.118
With PCA	0.394	0.367	0.360	0.337	0.333	0.342	0.356

4.3 Experiment III

Compare the silhouette score of K-means model with PCA on 740 samples and 527 samples to choose the better model to apply in the next experiment. From Table 4.3, the better silhouette score is K-means clustering due to the average and the highest silhouette score. The average silhouette of K-means clustering with PCA on 527 samples is 0.360, which isn't more than the average of 740 samples. But, the highest score is 0.405, with more than 740 samples around 3 percent.

Table 4.3 shows the silhouette score between the K-means model with PCA on 740 samples and 527 samples.

Cluster	3	4	5	6	7	8	Average
740 samples	0.394	0.367	0.360	0.337	0.333	0.342	0.356
527 samples	0.405	0.368	0.366	0.343	0.340	0.333	0.360

4.4 Result

4.4.1 Clustering result

The average score across all trials was 0.360. The clustering outcome with three clusters yields the highest score of 0.405. The following are characteristics of each cluster

Cluster 1: This cluster mainly consists of company employees and students aged around 21 to 40 with bachelor's degrees or lower education. Their income ranges from less than 15,000 to 30,000 Baht per month. They mostly own motorcycles but use private cars for travel. Their hometowns are mainly located in central Thailand. They are mostly single, with households containing two to three people. Their preference is for one to two-day trips taken less than twice a year. They express a desire to travel to the northern and central regions of Thailand, with a budget ranging from less than 500 to 2,500 Baht.

Cluster 2: This cluster mainly consists of company employees, including freelancers, government officers, business owners, and individuals with other occupations, aged between 31 to 60. Their education ranges from bachelor's degrees to higher-level degrees. The income typically falls within the range of 30,000 to more than 60,000 Baht per month. They mostly own private cars and prefer to use them for travel. Their hometowns are primarily located in central Thailand. The majority are married, and their households typically consist of four to five people. They express a desire to travel for three to four days per trip, with two to three trips per year. The preferred travel region is mainly central Thailand, with a budget of more than 3,500 Baht.

Cluster 3: This cluster mainly consists of students aged around 21 to 30 years old. Most have bachelor's degrees in education. Their income is mostly less than 15,000 Baht per month. They own motorcycles and other vehicles, but they tend to use private cars and airplanes for travel. Their hometowns are located in the south and north of Thailand. The majority are single, and households typically contain four or more people. They express a desire to travel for three to four days per trip, with two to more than five trips per year. The regions they want to explore are the north and south of Thailand, with a budget exceeding 3,500 Baht.

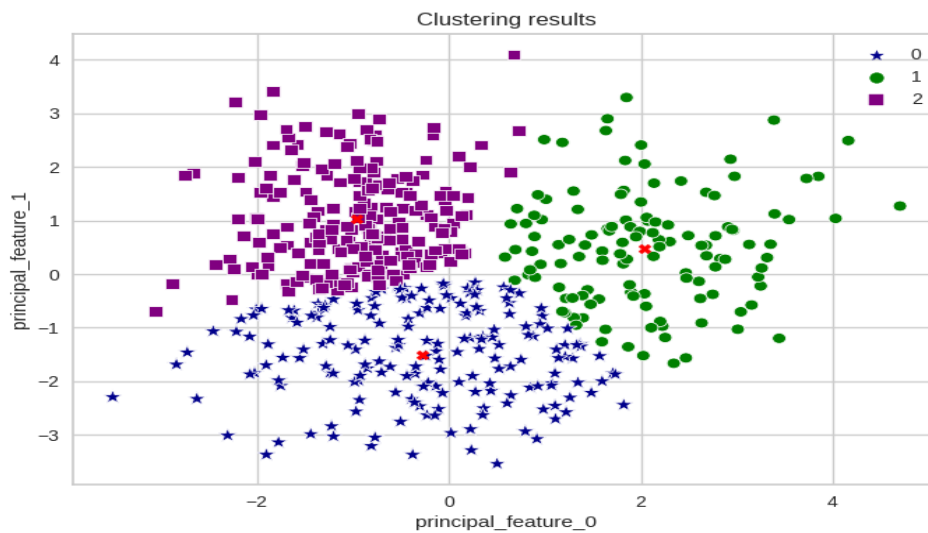


Figure 4.1 Shows the Clustering result

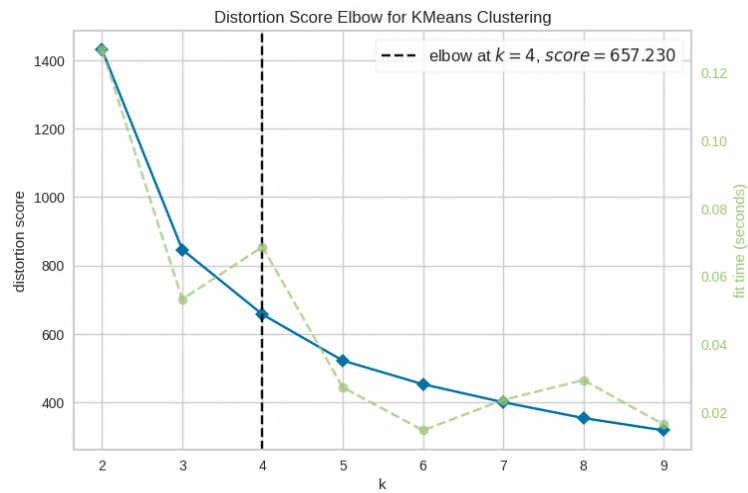


Figure 4.2 Shows the optimal k from the elbow method

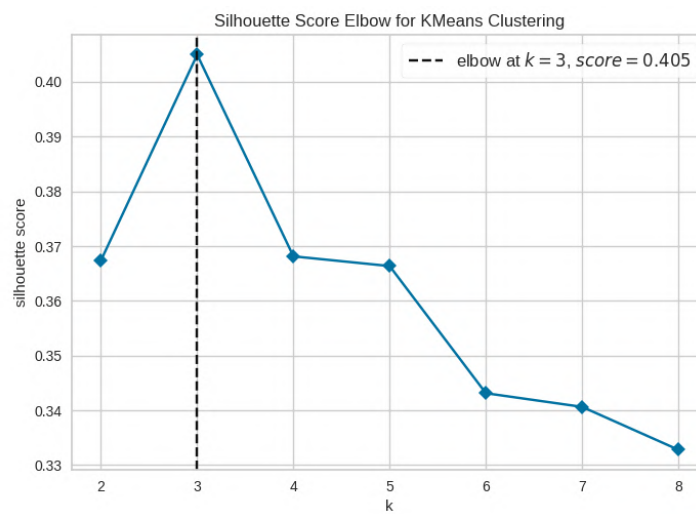


Figure 4.3 Shows the optimal k from the silhouette score

Table 4.4 Shows the distribution of the three clusters of travel behavior.

Attribute	Cluster			Total	Attribute	Cluster			Total
	1	2	3			1	2	3	
Gender					Status				
Alternative	3	1	12	16	Not mentioned	18	1	18	37
Male	42	41	35	118	Married	12	82	4	98
Female	143	83	166	393	Single	158	43	191	392
Age					Per_region				
<=20	18	0	35	53	Western	2	1	1	4
21-30	94	4	164	262	North-East	7	0	9	16
31-40	39	27	11	77	Eastern	4	2	15	21
41-50	31	65	3	98	South	17	10	49	76
51-60	6	24	0	30	North	22	29	51	102
>60	0	6	0	6	Central	136	84	88	308
Education					Freq				
Under bachelor's degree	75	27	27	129	Less than 2 times/year	90	44	33	167
Bachelor's degree	108	77	163	268	2-3 times/year	50	50	79	179
Higher bachelor's degree	5	22	3	30	4-5 times/year	19	17	45	81
Occupation					> 5 times/year	29	15	56	100
Government employee	1	0	0	1	Season				
Farmer	1	3	0	4	Rainy season	7	2	1	10
Housewife/husband	4	2	0	6	Hot season	41	31	41	113
Freelancer	8	10	3	21	Cold season	140	93	171	404
Government officers	2	18	6	26	Type				
Business owner	4	15	9	28	Business	3	0	1	4
Others	11	14	3	28	Ethnic	1	5	5	11
Company employees	97	63	26	186	Historical	9	7	3	19
Student	60	1	166	296	Cultural	19	5	7	31
Income					Recreational	24	2	29	55
<15000	84	16	152	252	Environment	132	107	168	407
15000-30000	82	40	57	179	Region				
30001-45000	21	35	4	60	Western	5	0	5	10
45001-60000	1	14	0	15	North-East	13	4	6	23
>60000	0	21	0	21	Eastern	29	11	9	49
Per_vehicle					South	28	41	62	131
None	5	5	7	17	North	54	54	108	216
Other	42	18	80	140	Central	59	16	23	98
Motorcycle	88	15	78	181	Person				
Private car	53	88	48	189	Single	21	2	11	34
Household					1-2	62	20	34	116
1	17	12	3	32	3-4	75	66	70	211
2-3	78	43	53	174	5	16	2	32	50
4-5	76	57	125	258	>5	14	36	66	116
>5	17	14	32	63	Expenditure				
Vehicle					<500	9	0	0	9
Passenger ship	1	0	0	1	501-1500	40	3	8	51
Train	5	2	8	15	1501-2500	46	6	18	70
Motorcycle	13	2	1	16	2501-3500	36	17	57	110
Public transport	27	8	14	49	>3500	57	100	130	277
Airplane	5	16	56	77	Period_time				
Private car	137	98	134	369	Weekday	12	21	44	77
Day_travel					Holidays	45	46	108	199
1	25	0	0	25	Weekend	131	59	61	251
1-2	140	42	40	222	Want_travel				
3-4	22	67	127	216	Definitely not!	7	1	2	10
>4	1	17	46	64	I'm not sure	74	27	42	143
					I'd love to	107	98	169	374

CHAPTER 5

CONCLUSION AND DISCUSSION

5.1 Conclusion

The objective of this project was to create a website utilizing machine learning-based clustering to comprehend the characteristics and preferred travel behavior of a sample of Thai people for domestic tourism post the COVID-19 pandemic. The goal is to suggest tour packages that align with user preferences. In the machine learning component, Principal Component Analysis (PCA) was applied to reduce the data's dimensionality, and K-means clustering was employed as the grouping algorithm. The optimal model configuration involved K-means clustering with PCA on 527 samples. Clustering the samples into three groups resulted in an optimal silhouette score of 0.405. The characteristics derived from this clustering will be leveraged to enhance the tourism industry by recommending tour packages for domestic travel.

5.2 Discussion

Due to the evaluation metric, we have chosen a k-cluster value of 3. However, the optimal k obtained from the elbow method is 4. The reason for selecting 3 clusters is based on a comparison with the silhouette coefficient score. Specifically, the silhouette score for 3 clusters is approximately 0.405, while the elbow method indicates a value of around 800. On the other hand, if we were to use 4 clusters, the silhouette score would decrease to 0.36, and the elbow method suggests a value of around 600. Comparing the transition from 2 clusters to 3 clusters, the elbow method decreases from 1400 to 800. Consequently, we have decided to proceed with 3 clusters.

Key successes of the project include the seamless integration of the machine learning model with the web application and the intuitive UX/UI design that enhances user interaction. Challenges such as ensuring responsive design and integrating complex datasets were effectively addressed. The use of modern web technologies and frameworks has allowed for scalable and maintainable web architecture. Future enhancements could include expanding the recommendation engine's capabilities and incorporating real-time data for more dynamic travel suggestions. The project's impact lies in its innovative approach to personalized travel planning, setting a new standard for travel websites.

Another possibility is to enhance clustering performance to ensure higher accuracy in matching the right touring package to a customer.

5.3 Future work

1. To create a more precise model, gather additional data and enhance the clustering method by experimenting with new models and adjusting parameters.
2. Improve the security, design, and performance of the website.
3. Launch the website dedicated to the economic sector, aiming to be a pivotal element in enhancing the tourism industry in Thailand.

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